

CSE5026: Foundation and Frontiers of Cognitive Science

Vision

Yang Xu 徐炆

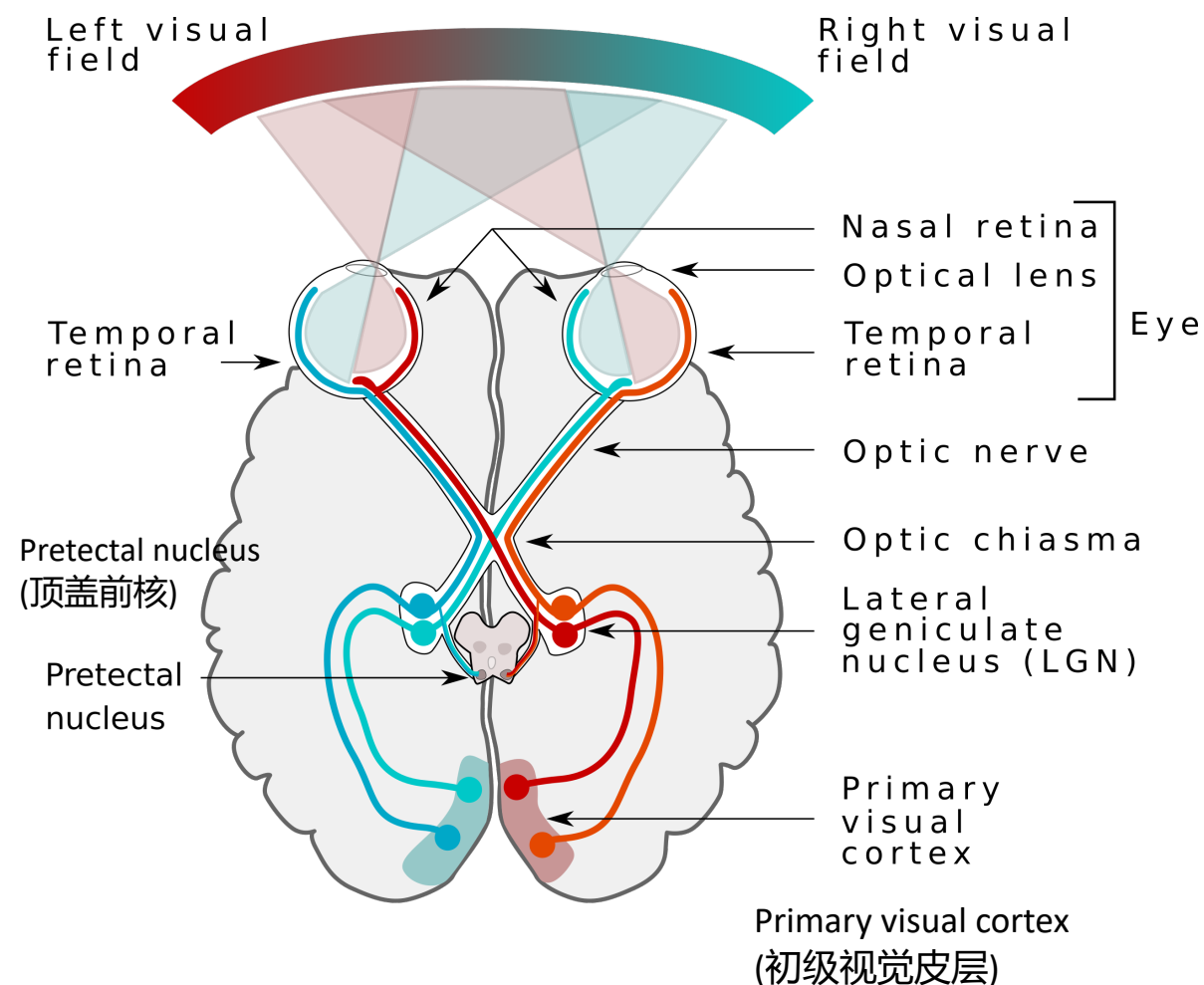
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Overview

- Physiological Basis of Vision
- Visual Cortex (视觉皮层)
- Theories about Vision
- Convolutional Neural Networks

Physiological Basis of Vision

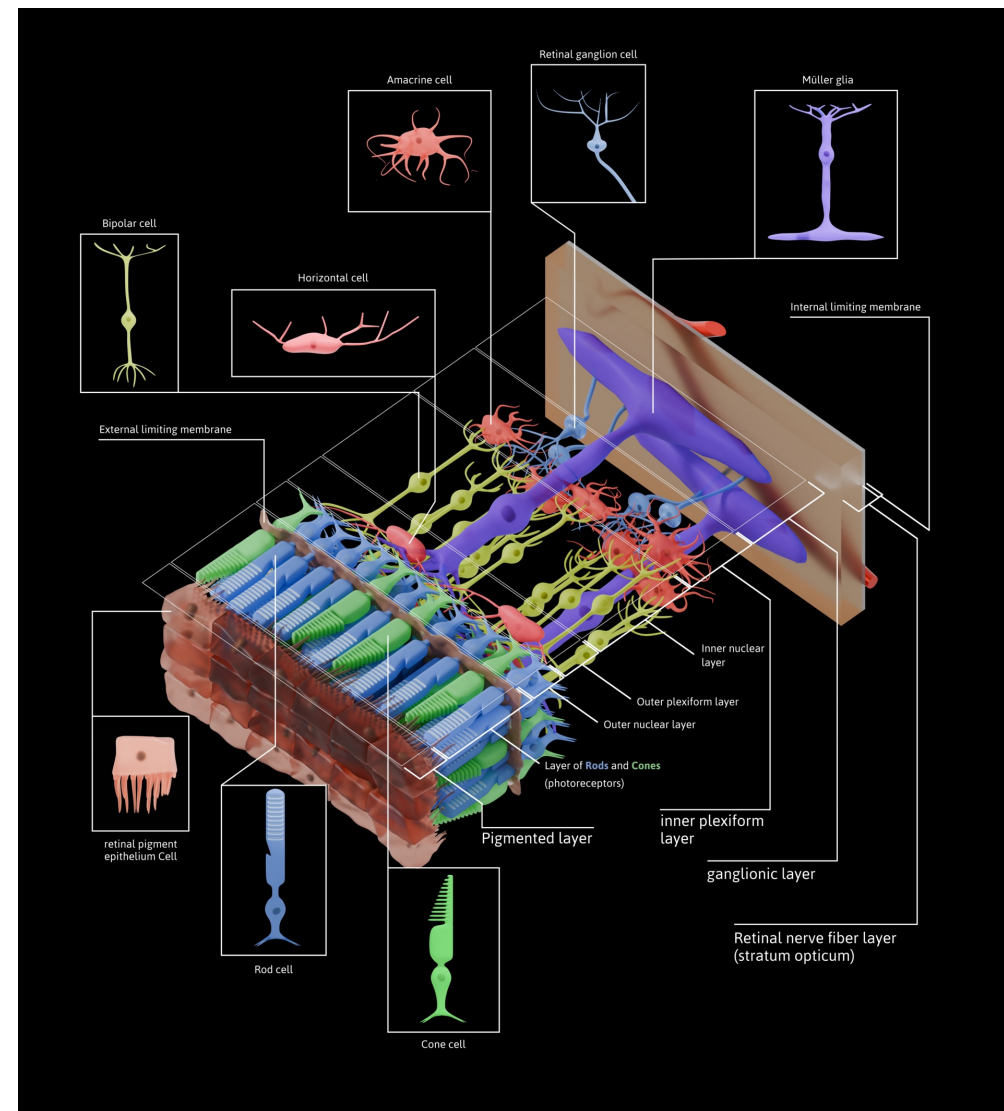
- Vision provides much of our information about the environment.
- The eye is a highly complex sensory organ.
- Light enters the eye and forms an image on the retina (视网膜).
- Signals travel through the optic nerve (视神经) to visual areas of the brain.



Source: https://en.wikipedia.org/wiki/Visual_system

Retina (视网膜)

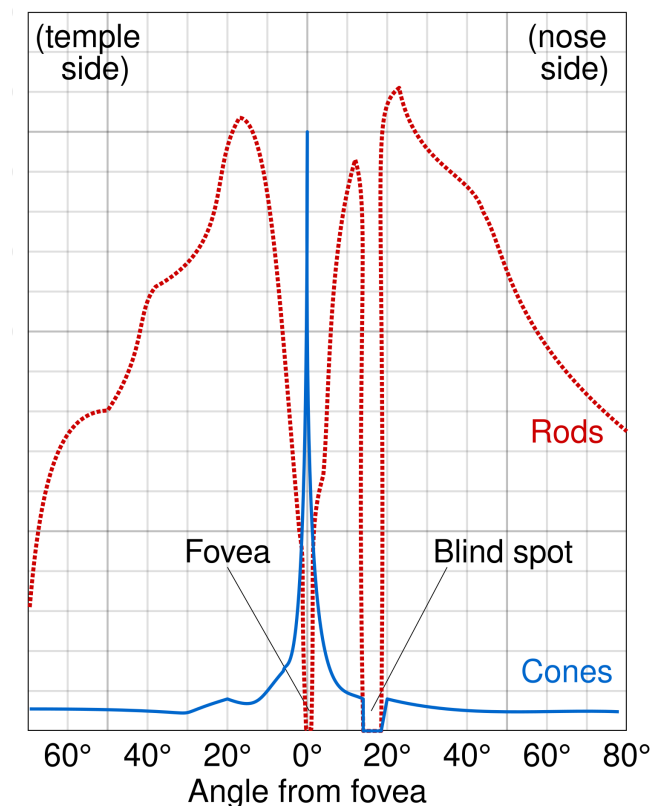
- The retina is about 0.5 mm thick.
It is part of the brain.
- Three stages of neurons (神经元):
 - 1. Photoreceptors (光感受器)
 - **Rods** (视杆细胞) and **cones** (视锥细胞)
 - 2. Bipolar cells (双极细胞)
 - 3. Ganglion cells (神经节细胞)
 - Their axons (轴突) form the optic nerve (视神经).



Source: https://en.wikipedia.org/wiki/Visual_system

Retinal Cells

- 柱细胞 • **Rods** detect very dim light but do not distinguish colors. About 120 million per retina.
- 锥细胞 • **Cones** support color vision in brighter light. About 6 million per retina.
- Ganglion cells are much fewer in number.
 - Many photoreceptors converge through bipolar cells onto a single ganglion cell.
 - Cone density peaks in the fovea (中央凹), where visual acuity is highest. (视敏度)

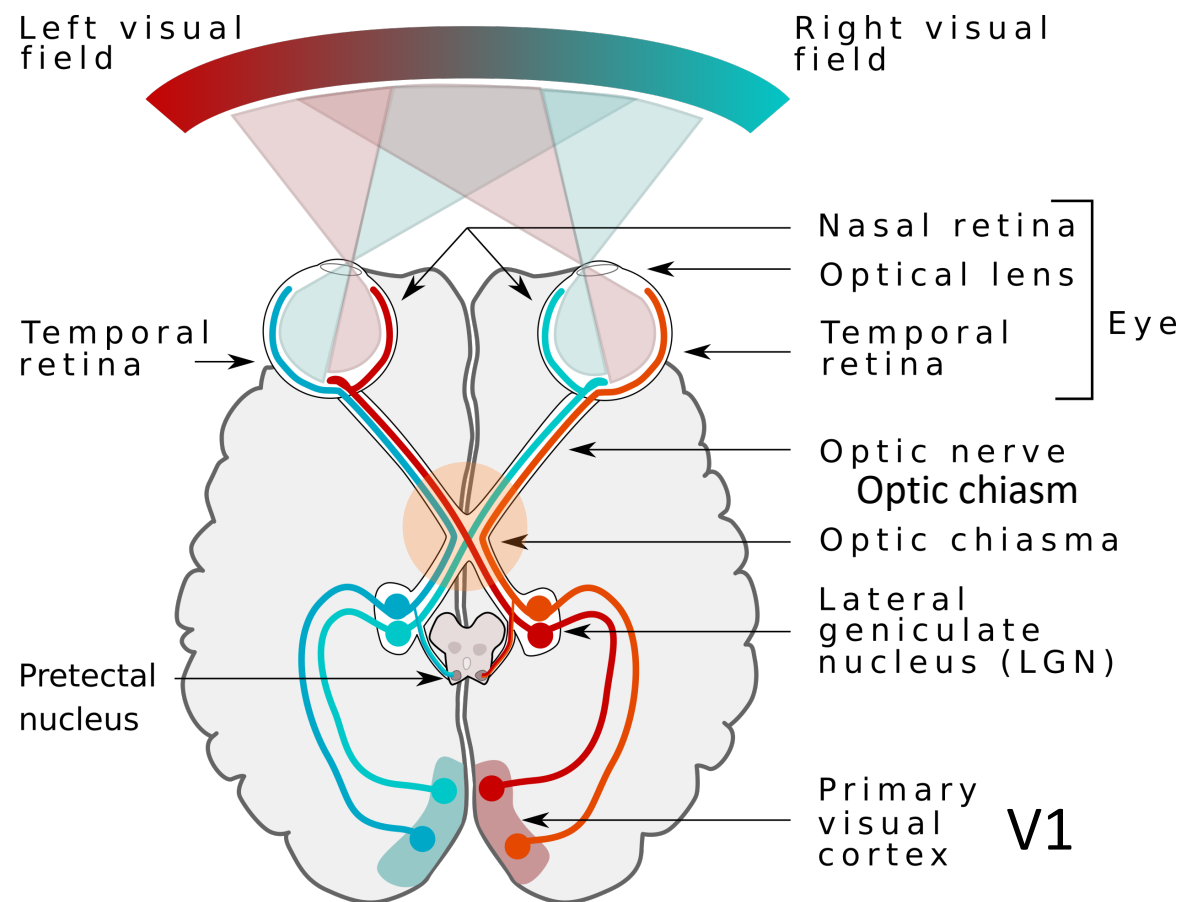


Source: Webvision, Facts and Figures Concerning the Human Retina

Visual Pathway (视觉神经通路)

[ˈkaɪæzəm]

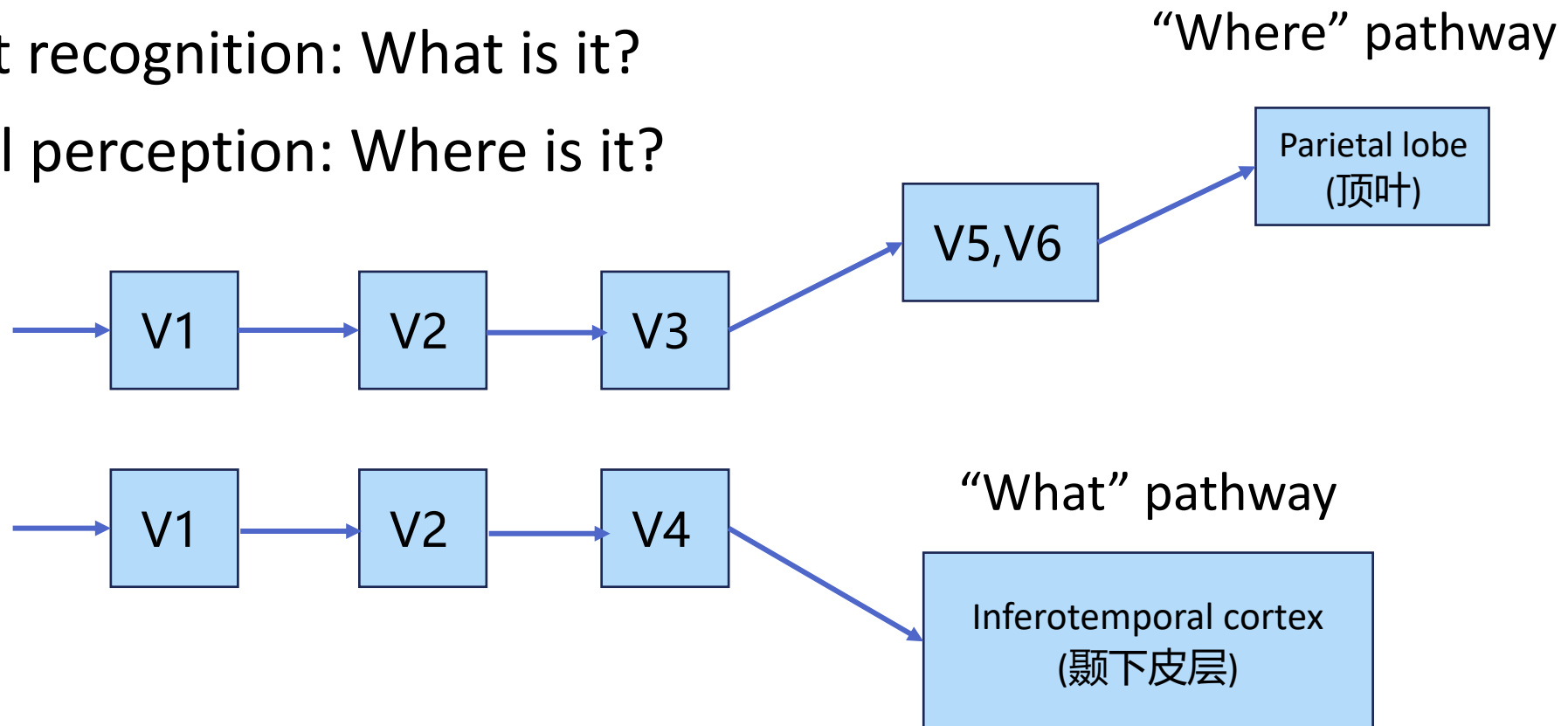
- The optic nerves meet at the optic **chiasm** (视交叉), at the base of the brain.
- Some fibers cross, routing each visual field to the *opposite* hemisphere. (脑半球)
 - Right visual field: left hemisphere.
 - Left visual field: right hemisphere.
- The *lateral geniculate nucleus* (外侧膝状体核, LGN) has six layers that relay and process visual signals.
- Signals then reach primary visual cortex (V1).



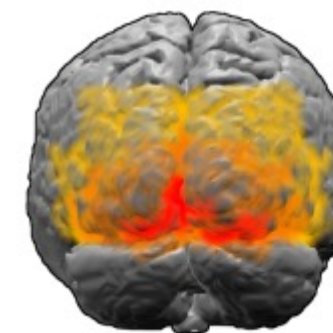
Source: https://en.wikipedia.org/wiki/Visual_system

Visual Pathway

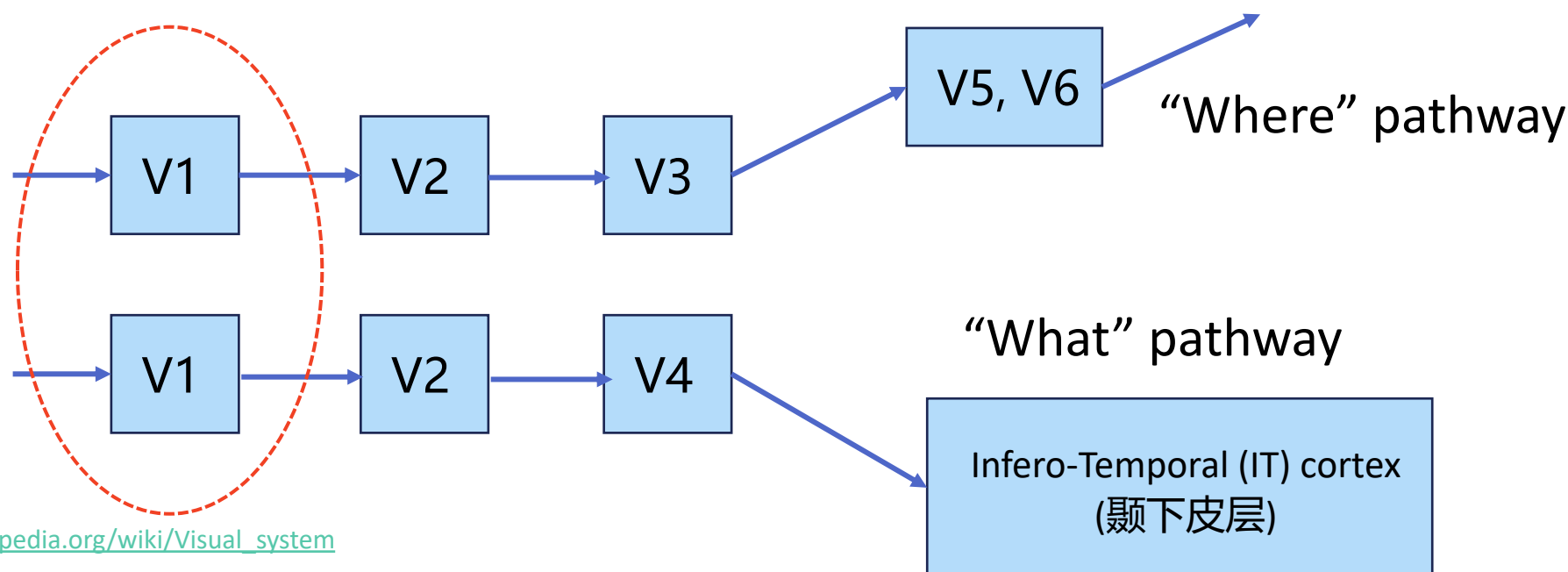
- Two main functions of vision:
 - Object recognition: What is it?
 - Spatial perception: Where is it?



Visual Cortex

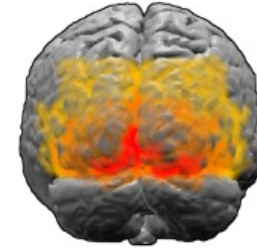


- Visual cortex lies in the occipital lobe (枕叶).
- Primary visual cortex (**V1**) receives input directly from the LGN.
- The **V1 saliency hypothesis (V1SH)** proposes that V1 builds a bottom-up saliency map to guide attention and gaze. 显著性

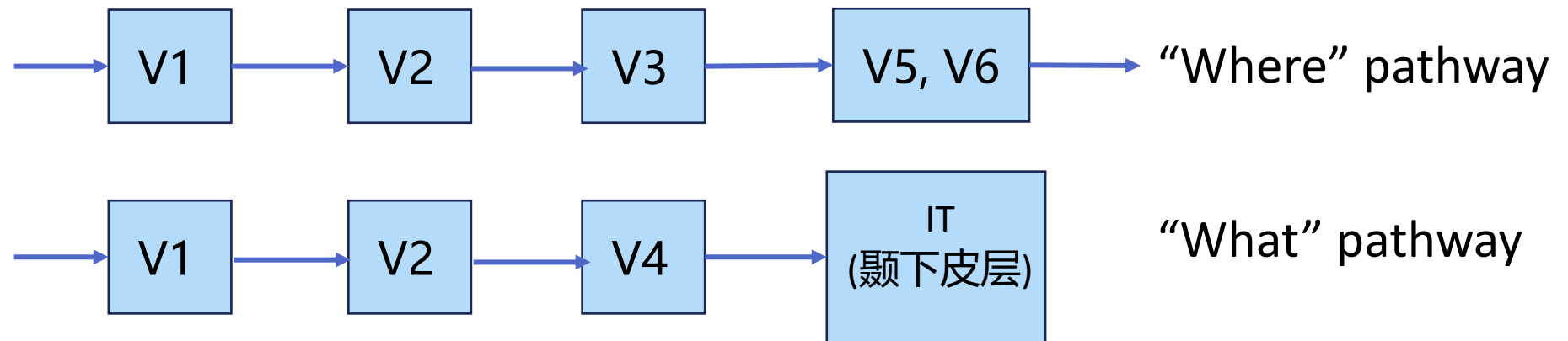


Source: https://en.wikipedia.org/wiki/Visual_system

Visual Cortex



- Visual processing extends across V2, V3, V4, V5 (MT) and V6.
 - V1 and V2 respond to oriented *lines* and support *edge* and corner detection, as well as color and motion processing.
 - V3 contributes to *motion* processing. V4 contributes to *shape* processing.
 - V5 (MT) processes motion. V6 processes motion relative to the background.



Source: https://en.wikipedia.org/wiki/Visual_system

Visual Cortex

Source:

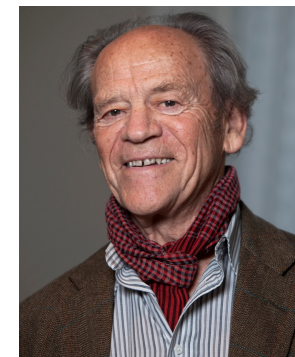
https://en.wikipedia.org/wiki/Simple_cell

https://en.wikipedia.org/wiki/Complex_cell

- Hubel and Wiesel (1959, 1962) studied neuronal firing in the cat's visual cortex.
- Their studies distinguished simple cells (简单细胞) and complex cells (复杂细胞). Later work described hypercomplex cells (超复杂细胞, 1965).

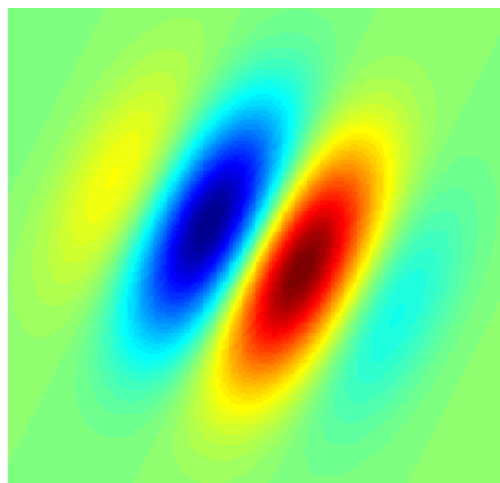


David H. Hubel



Torsten Wiesel

Simple cells in V1 respond to edges and gratings at particular orientations.



Complex cells occur in V1, V2 and V3.

They respond to oriented features, but tolerate some changes in position.

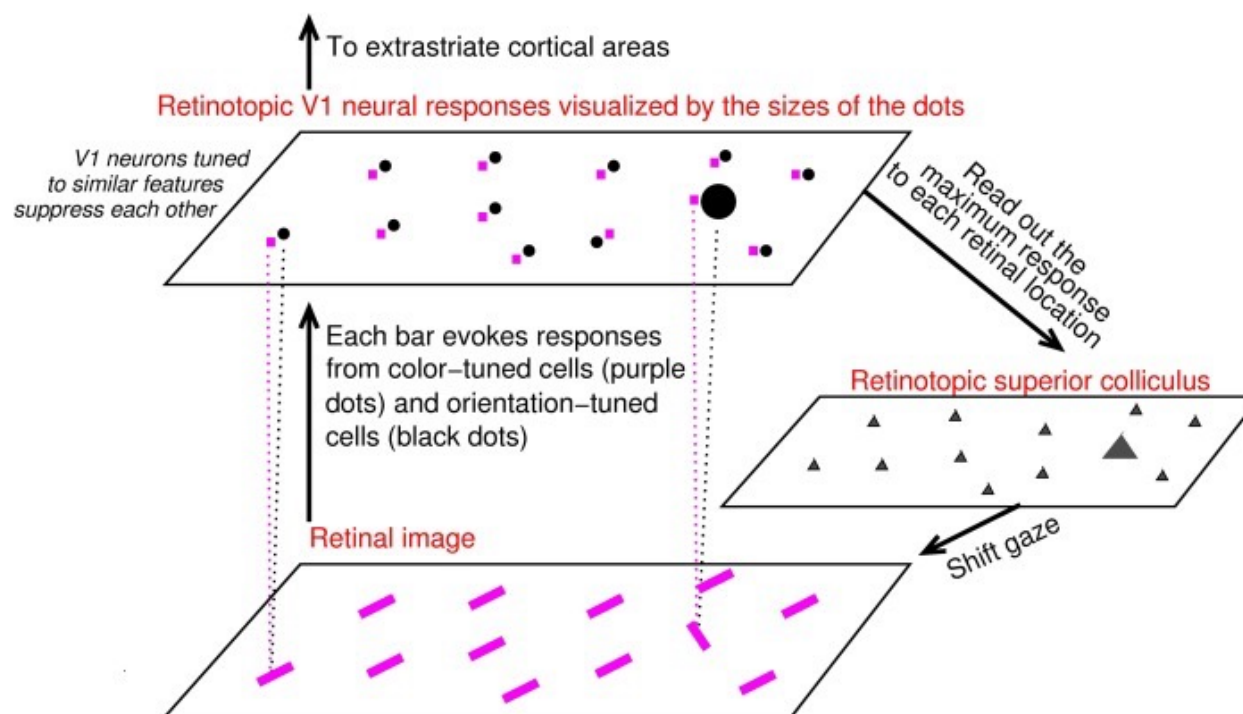
A Gabor filter model of a simple cell's receptive field (感受野)



Zhaoping Li
李兆平

V1 Saliency Hypothesis (V1SH)

- The hypothesis proposes that primate V1 creates a saliency map that guides visual attention and shifts of eye-gaze.



- V1 neurons prefer particular orientations.
- Iso-orientation suppression (同向抑制): nearby neurons with similar preferences suppress each other.
- An orientation singleton escapes suppression, giving a stronger response (largest black circle).
- The superior colliculus (上丘) in the midbrain (中脑) receives these signals and directs gaze shifts.

Source: https://en.wikipedia.org/wiki/V1_Saliency_Hypothesis

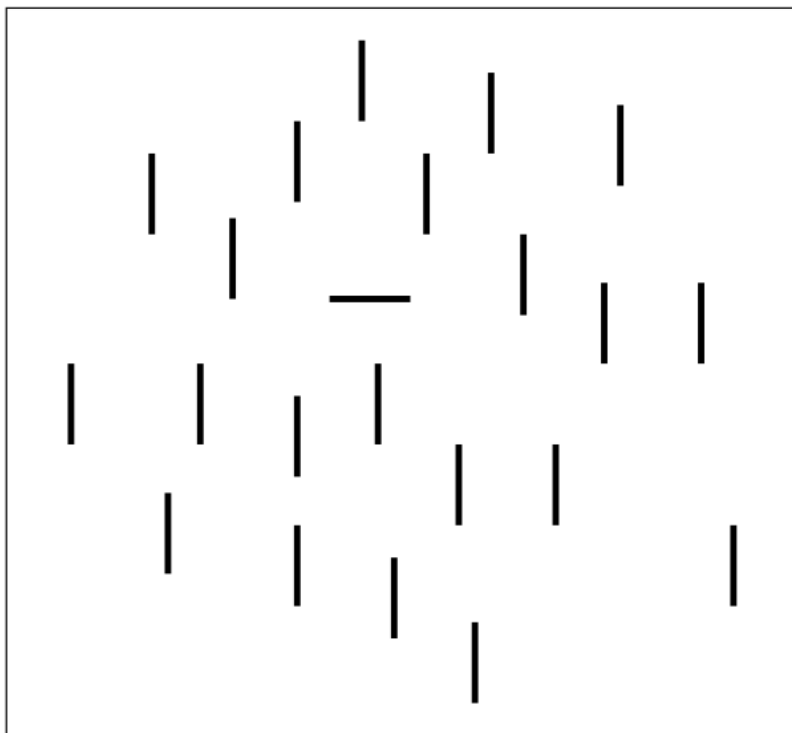
Zhaoping Li, 2002, [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Visual search tasks

(视觉搜索任务)

(a) Feature search

One feature: orientation



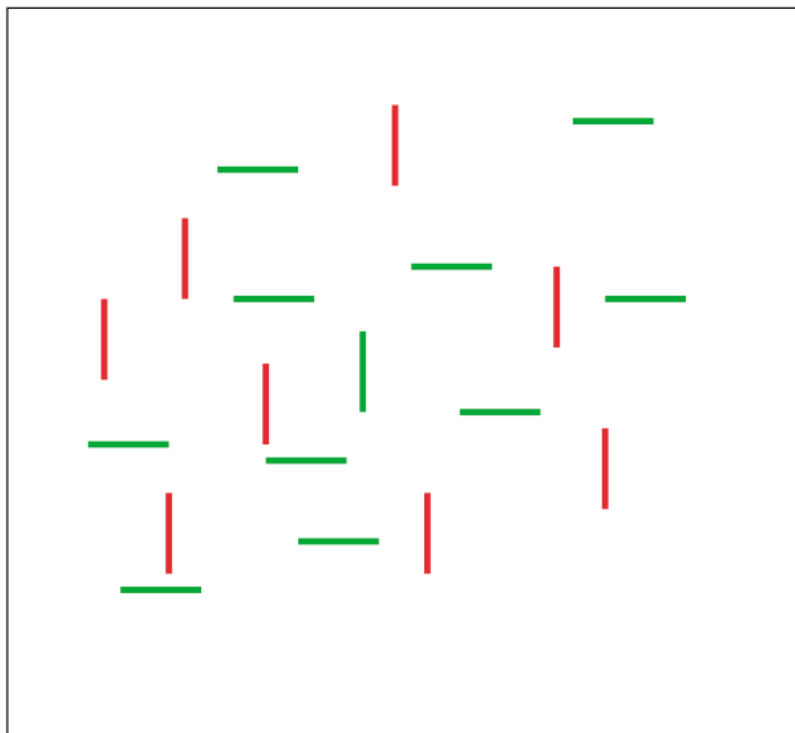
Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Visual search tasks

(视觉搜索任务)

(b) Conjunction search

Two feature: orientation + color

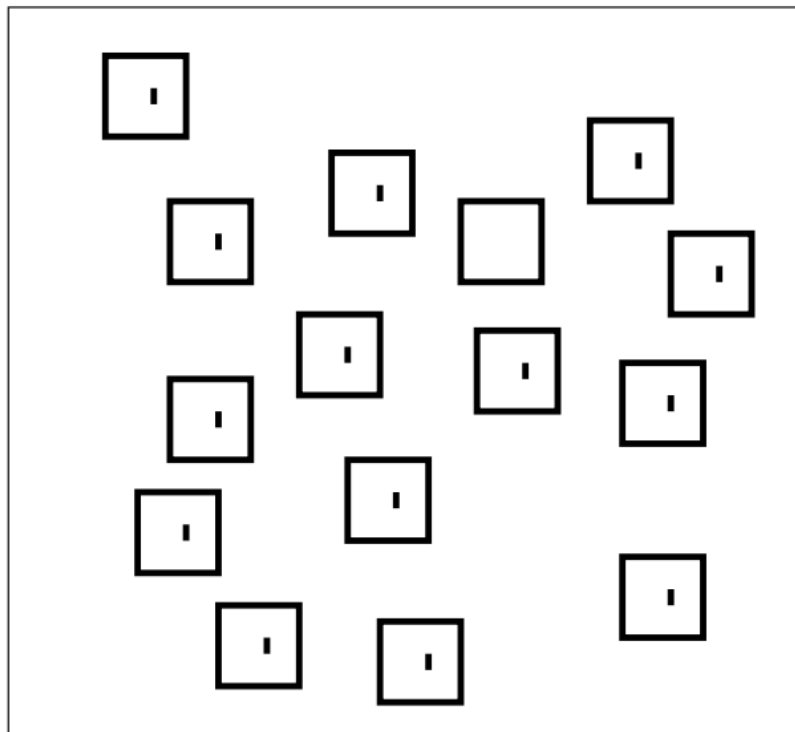


Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Visual search tasks

(视觉搜索任务)

(c) Target lacking a feature



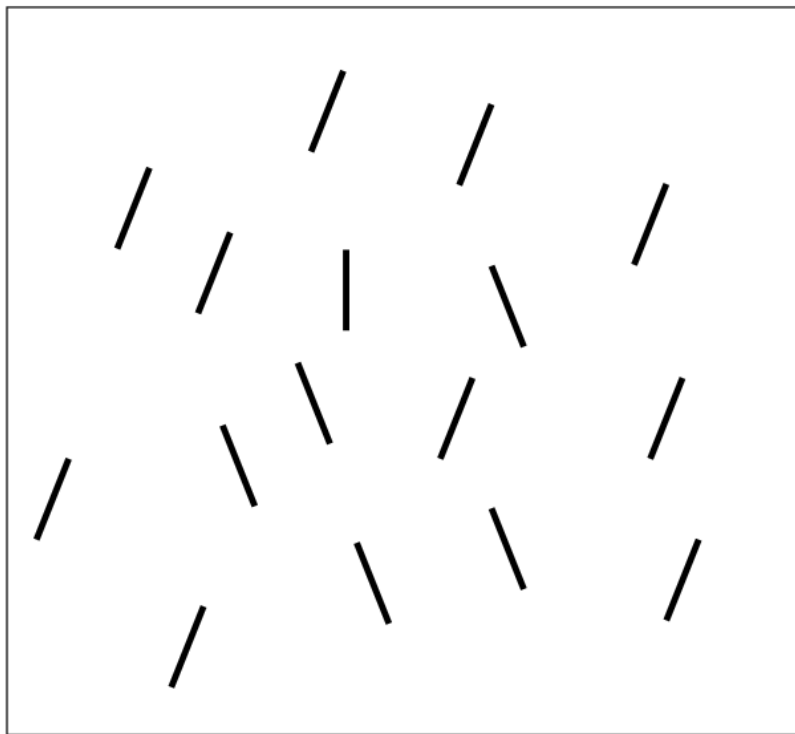
Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Visual search tasks

(视觉搜索任务)

(d) Dissimilar distractors

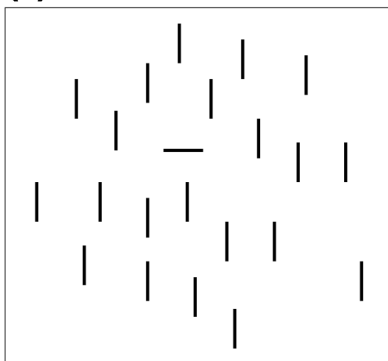
Two different kinds of distractors



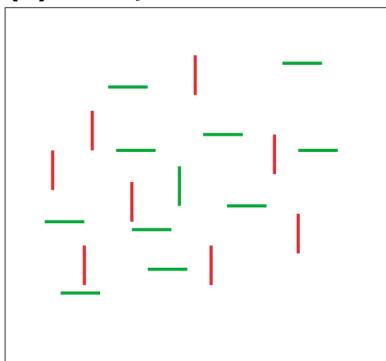
Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Visual search tasks (视觉搜索任务)

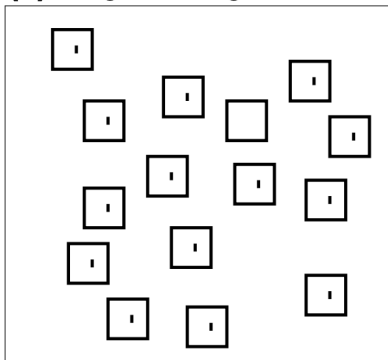
(a) Feature search



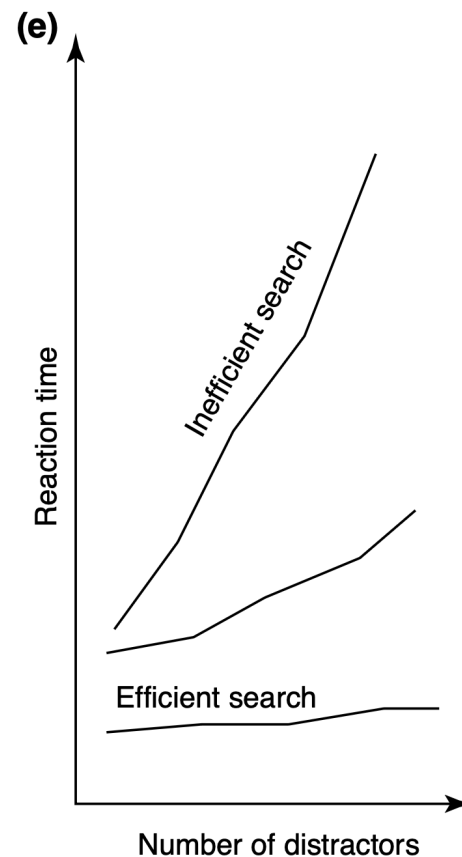
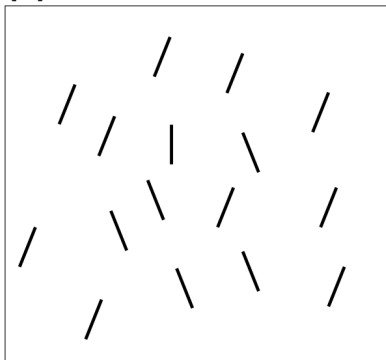
(b) Conjunction search



(c) Target lacking a feature



(d) Dissimilar distractors



In a visual search task, observers look for a target among distractors and report whether it is present. Researchers measure response time and accuracy.

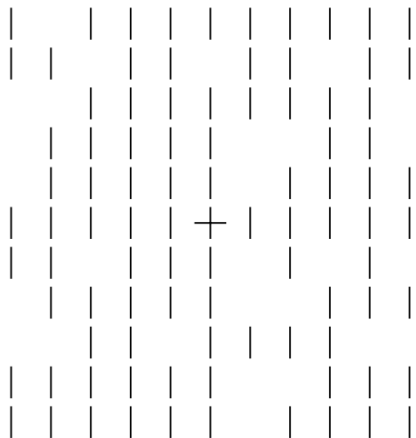
- Search efficiency is measured by the slope of reaction time (RT) versus the number of distractors.
- A steeper slope indicates less efficient search.

Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

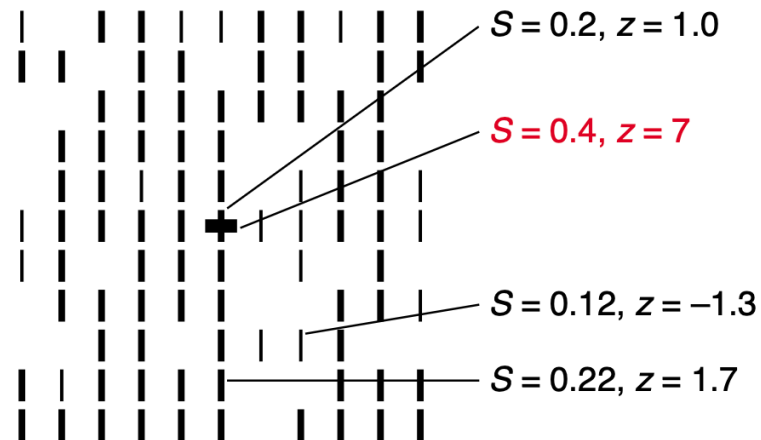
V1's output as a saliency map

Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

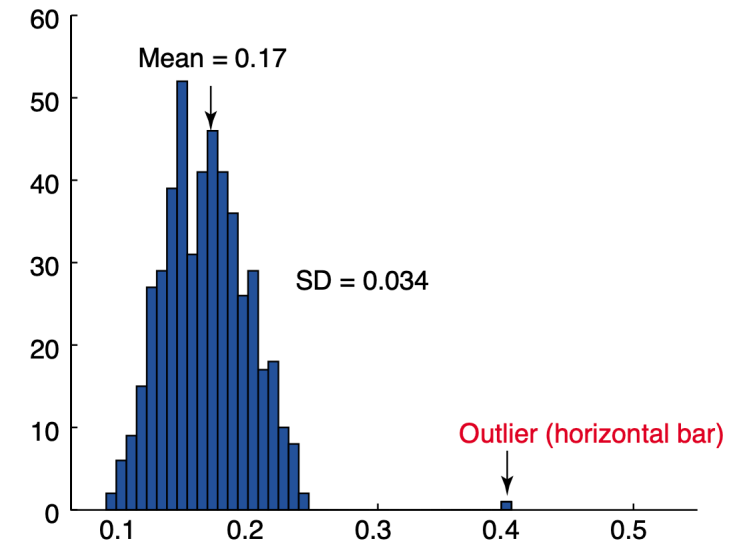
- Given a cell response, s_i , compute its saliency order $z_i = \frac{s_i - \bar{s}}{\sigma(s)}$



Visual input

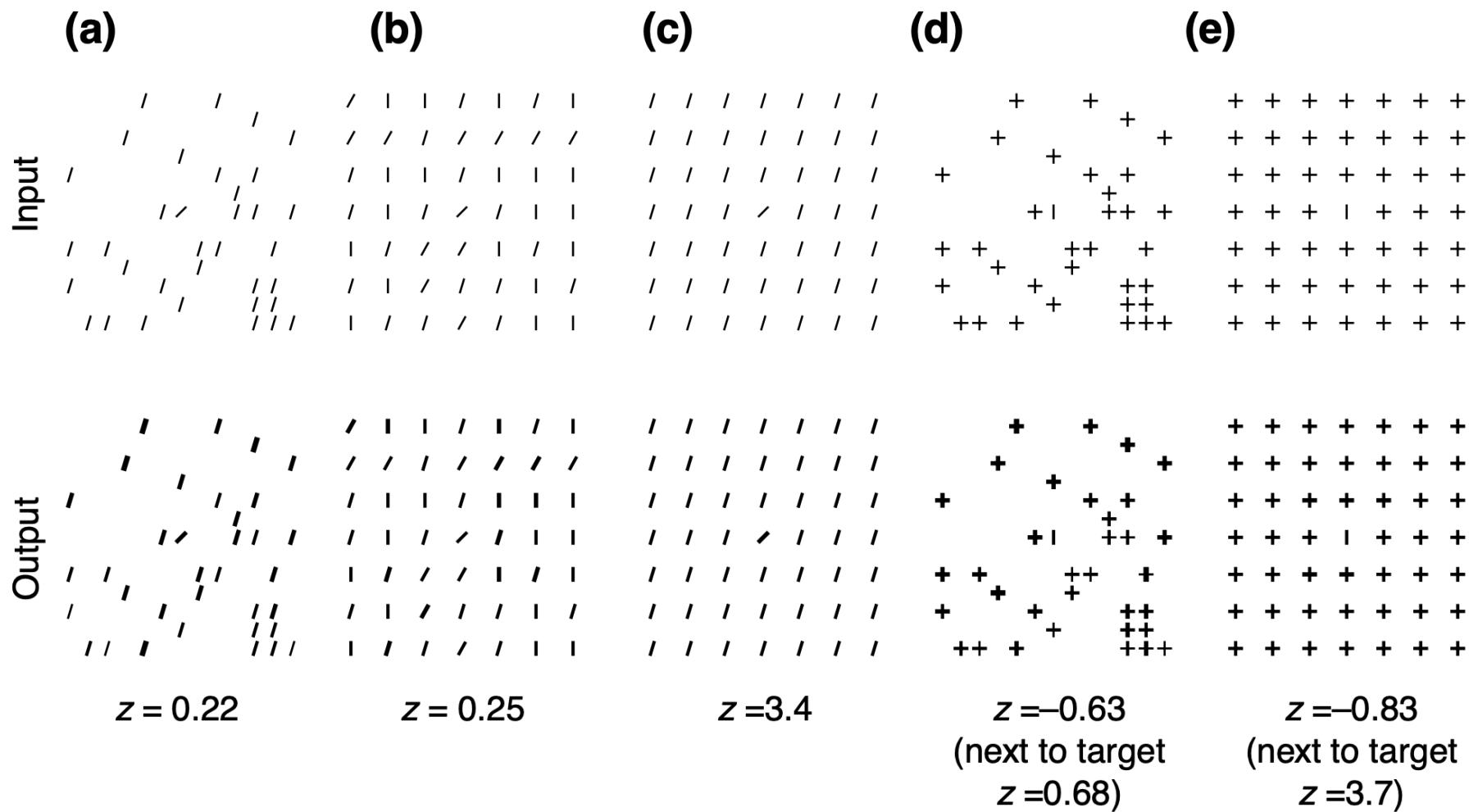


V1's output



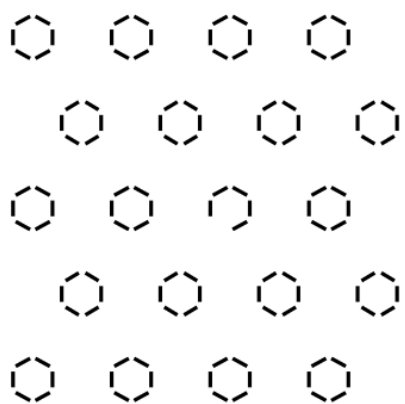
Histogram of s_i

Effect of background homogeneity



Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

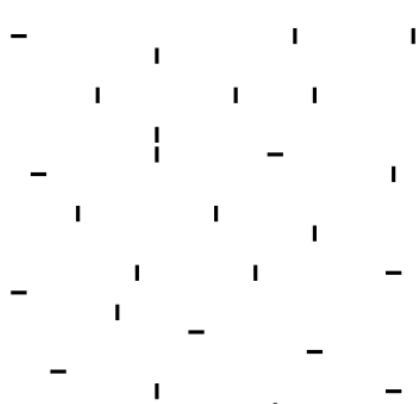
Asymmetries in visual search



$z = 9.7$



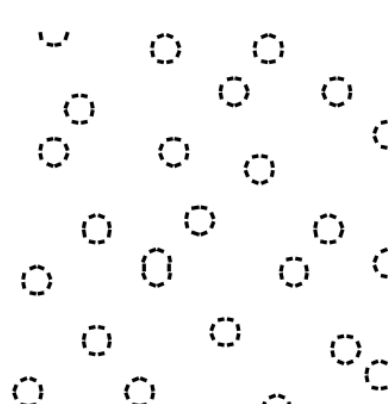
$z = 1.8$



$z = 1.07$



$z = 1.12$

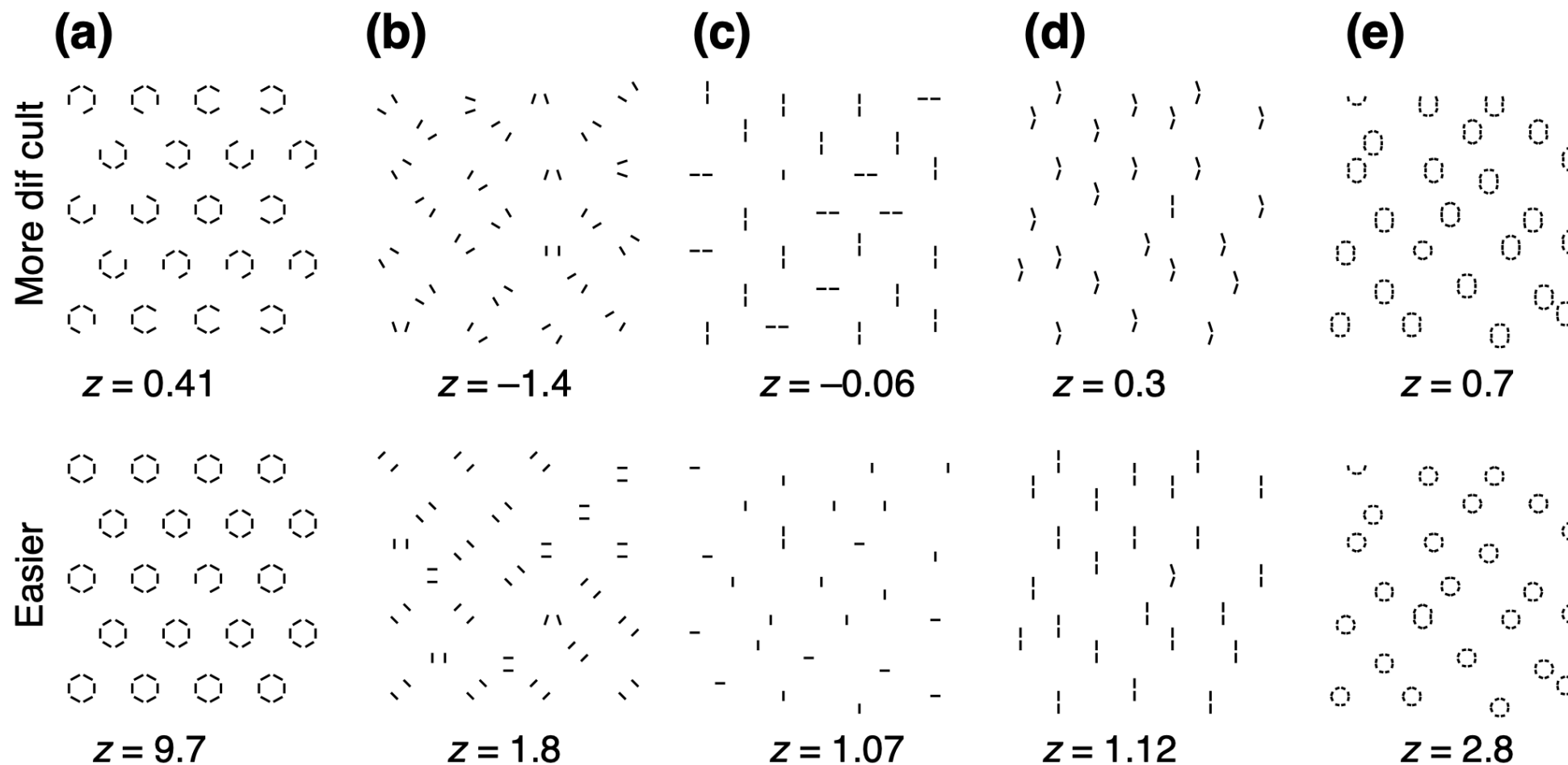


$z = 2.8$

Relatively easy

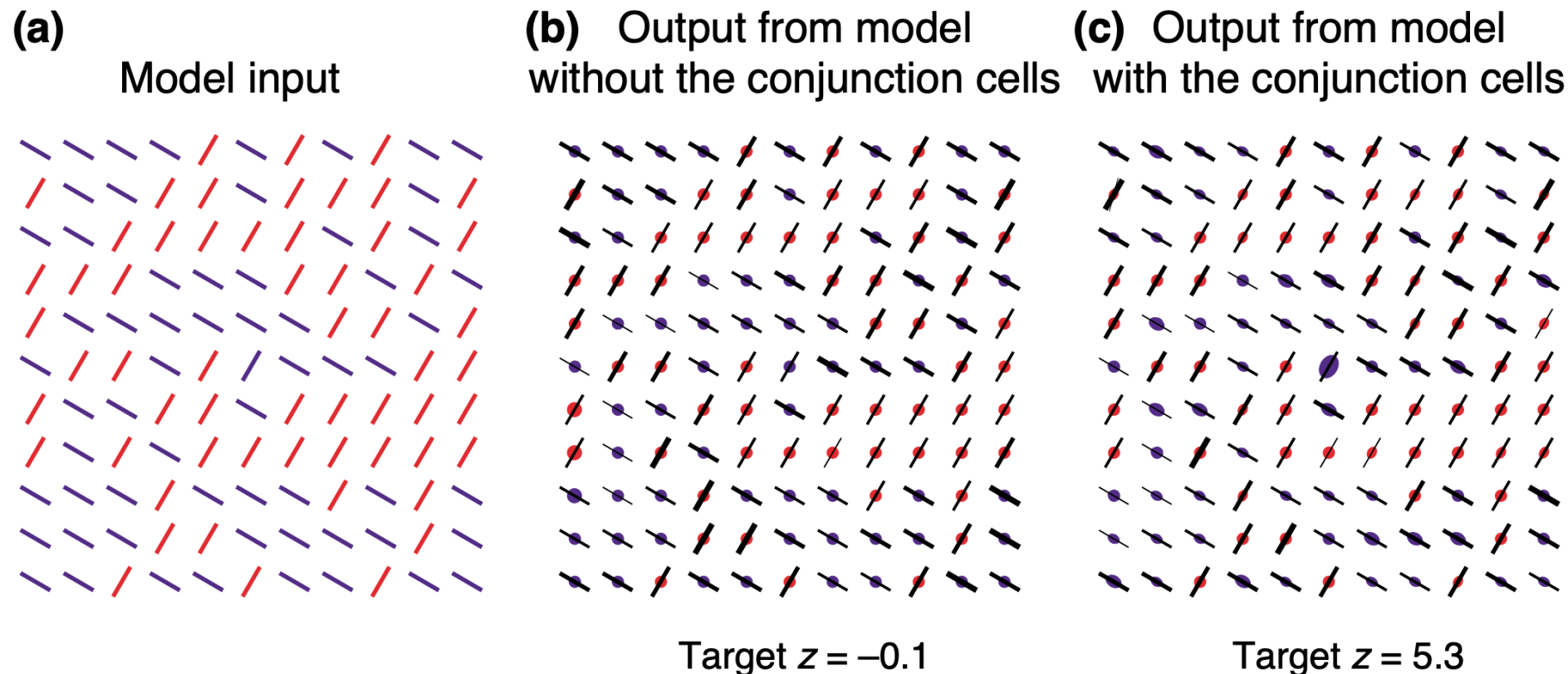
Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Asymmetries in visual search



Source: [doi: 10.1016/S1364-6613\(00\)01817-9](https://doi.org/10.1016/S1364-6613(00)01817-9)

Conjunction searches are more difficult

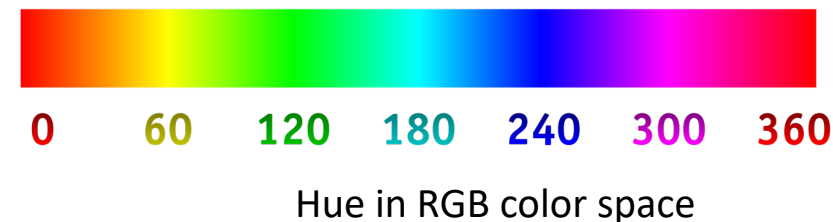


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Overview

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- **Theories about Vision**
- Convolutional Neural Networks

Color Theory: Color Attributes



(非彩色)

- Achromatic colors: black, white and shades of gray.

(彩色)

- Chromatic colors have three attributes:

色调 • **Hue**: the perceived color, such as red or green.

明度 • **Brightness**: how light or dark a color appears.

饱和度 • **Saturation**: color purity. Adding white reduces saturation.

Brightness and Saturation

Brightness (明度)

How light or dark
the image appears.

Lower brightness



Higher brightness



Saturation (饱和度)

How vivid or muted
the colors appear.

Lower saturation



Higher saturation

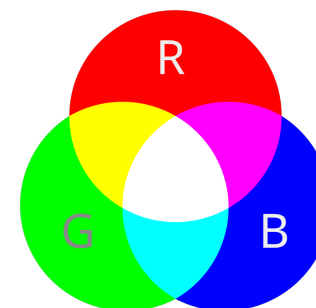


Color Theory: Color Mixing

- Three primary colors can mix to produce many colors.

- Additive mixing: **RGB** (加色混合)

- Red, green and blue light combine by adding their spectra.



- Subtractive mixing: **CMYK** (减色混合)

- Inks absorb parts of white light: cyan, (青色) magenta, yellow and black (K).

[mə'jen(t)ə] (品红)

白 $W = R + G + B$ (Everything)

青 $C = W - R = G + B$
(Red absorbed)

品红 $M = W - G = R + B$
(Green absorbed)

黄 $Y = W - B = R + G$
(Blue absorbed)



Sources:

https://en.wikipedia.org/wiki/RGB_color_model

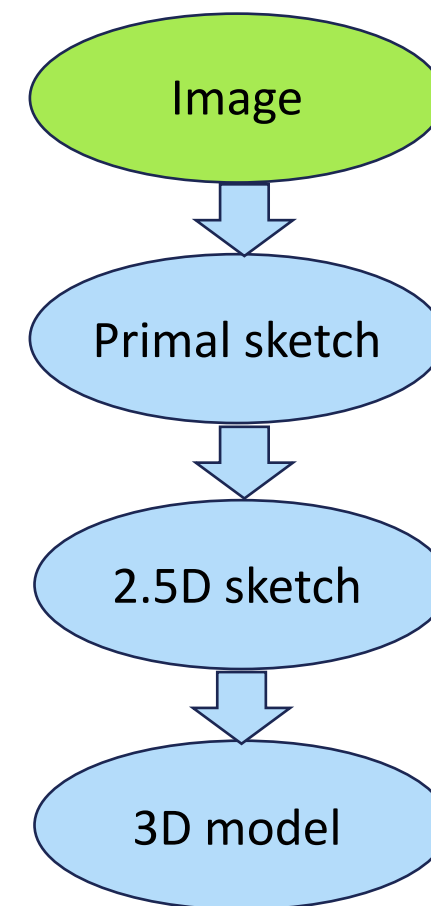
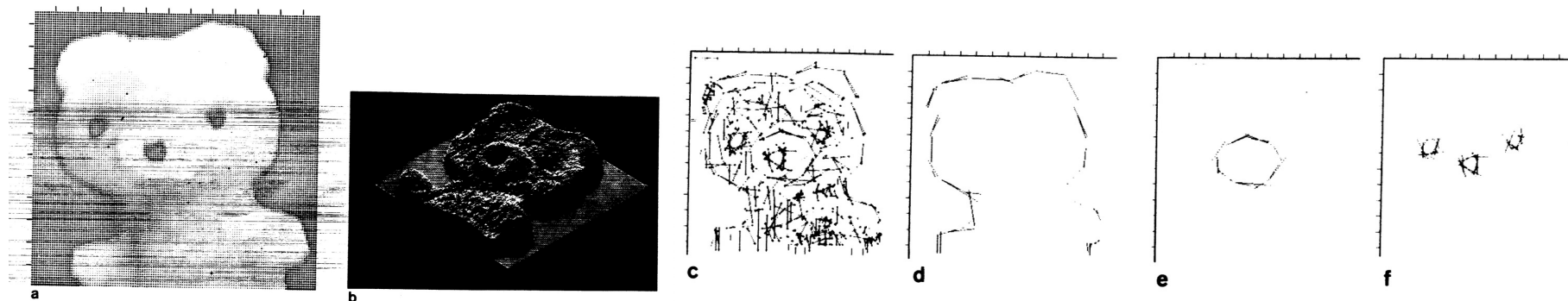
https://en.wikipedia.org/wiki/CMYK_color_model

Marr's Theory

David Marr (1945-1980), British cognitive scientist

- Visual processing builds three levels of representation.
- **Primal sketch**: basic features such as edges and regions, like a quick pencil sketch.

(原始草图)



Source: <https://dspace.mit.edu/handle/1721.1/5782>

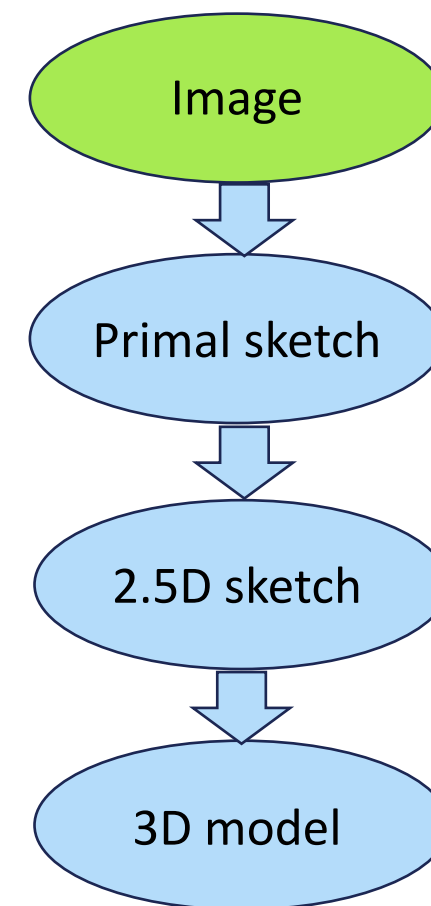
Marr's Theory

- **2.5D sketch:** a richer description of visible surfaces.
 - Represents surface orientation and depth, cues from disparity, lighting, texture and materials. (双眼视差)
- It bridges the primal sketch and the 3D model.



Disparity cues produce stereopsis (立体视觉), a sense of depth.

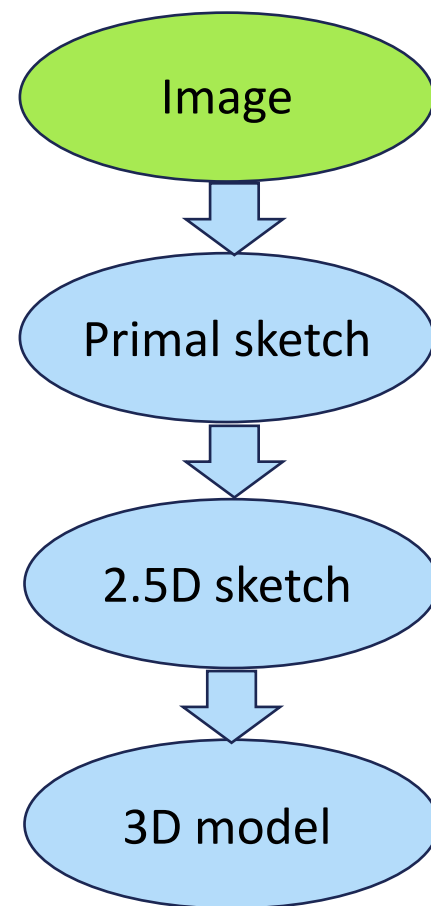
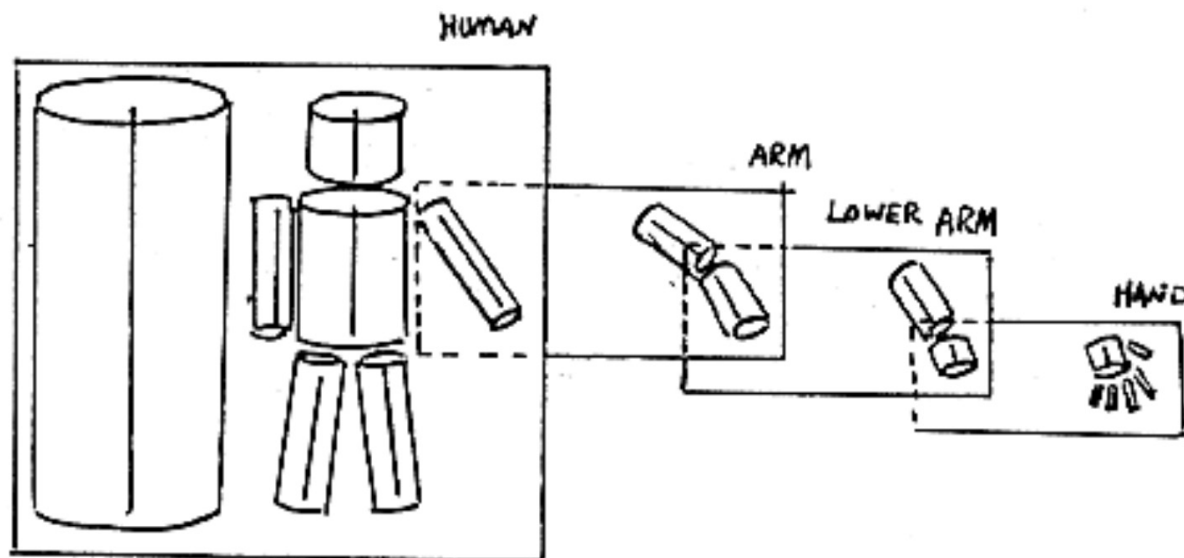
Source: <https://en.wikipedia.org/wiki/Stereopsis>



Marr's Theory

主要坐标轴

- A **3D model** decomposes a shape around its principal axes.
- Each axis gives the approximate spatial arrangement of a part.
- The axes organize the object's major components.



Topological Theory of Vision

Lin Chen 陈霖
CAS Fellow

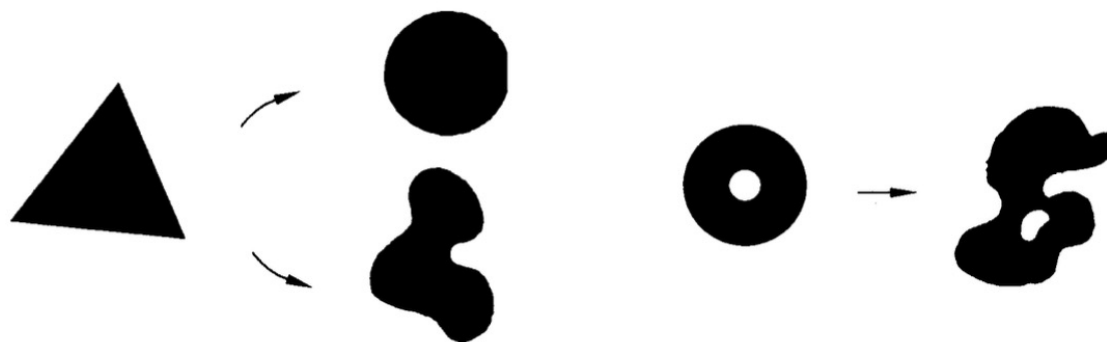


(原子主义) (整体主义)

- **Atomism** and **holism** offer contrasting accounts of visual perception.
 - Atomism: perception starts with features or simple parts, progressing from local to global properties.
 - Holism: perception starts with the whole, progressing from global to local properties.
 - Marr's theory follows the first approach.
- Lin Chen's (1982) early topological perception hypothesis follows the second.
 - The visual system detects global topological properties at the earliest stages of processing.

Topological Transformations and Properties

- Topology studies properties and relations preserved under continuous deformation.
(连接) (相邻) (包含)
- These include connectedness, adjacency and containment.
(Imagine stretching a rubber sheet without cutting it or gluing points together.)



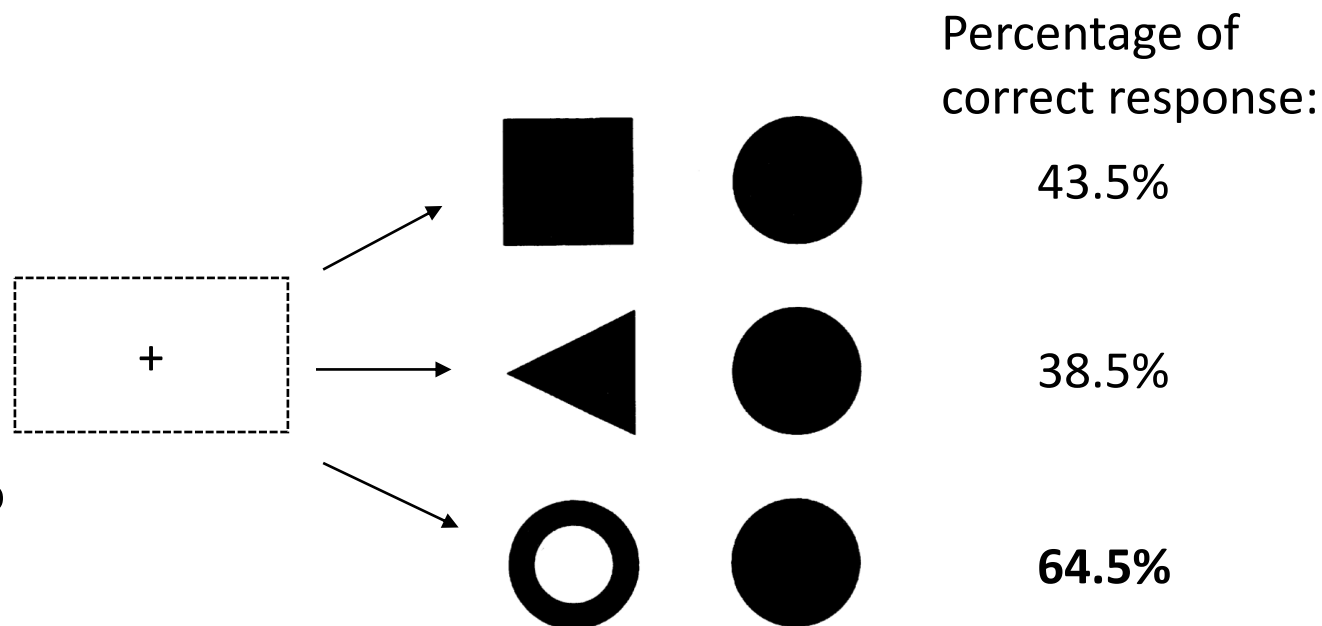
Local Features or Topology?

Lin Chen 陈霖, 1982

- A disk usually looks more similar to a ring than to a triangle or square.
- If topology is available early, very brief displays (~5 ms) may allow us to distinguish a disk from a ring even when we cannot distinguish it from a triangle or square.

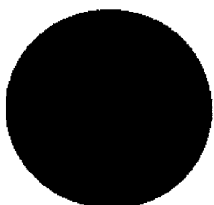
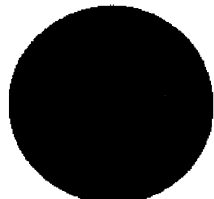
Procedure:

- Fixate the central cross.
- View each display for 5ms.
- An immediate blank mask disrupts perception.
- Judge whether the displayed two shapes are same or different.



Source: [doi:10.1080/13506280444000256](https://doi.org/10.1080/13506280444000256)

S-like Figure and Ring

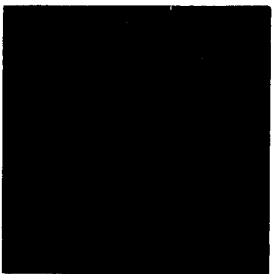
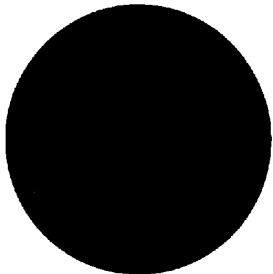
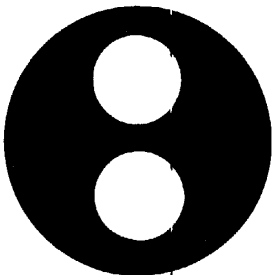
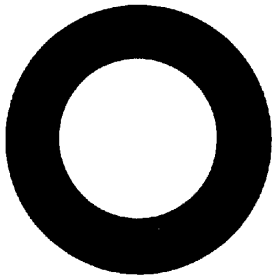
		% correct response	Area difference (mm ²)
		35.6	310
		59.0	0
		41.1	314

Hole difference: 0 vs. 1

The S-like figure and ring have equal area and similar local features.

They are easier to distinguish than the pairs with the same number of holes.

One Hole vs. Two Holes

		% correct response	Area difference (mm ²)
		39.0	310
		66.2	0

Hole difference: 2 vs. 1

The two small holes have the same total area as the large hole.

The hole-number difference is easier to detect than the solid-shape difference.

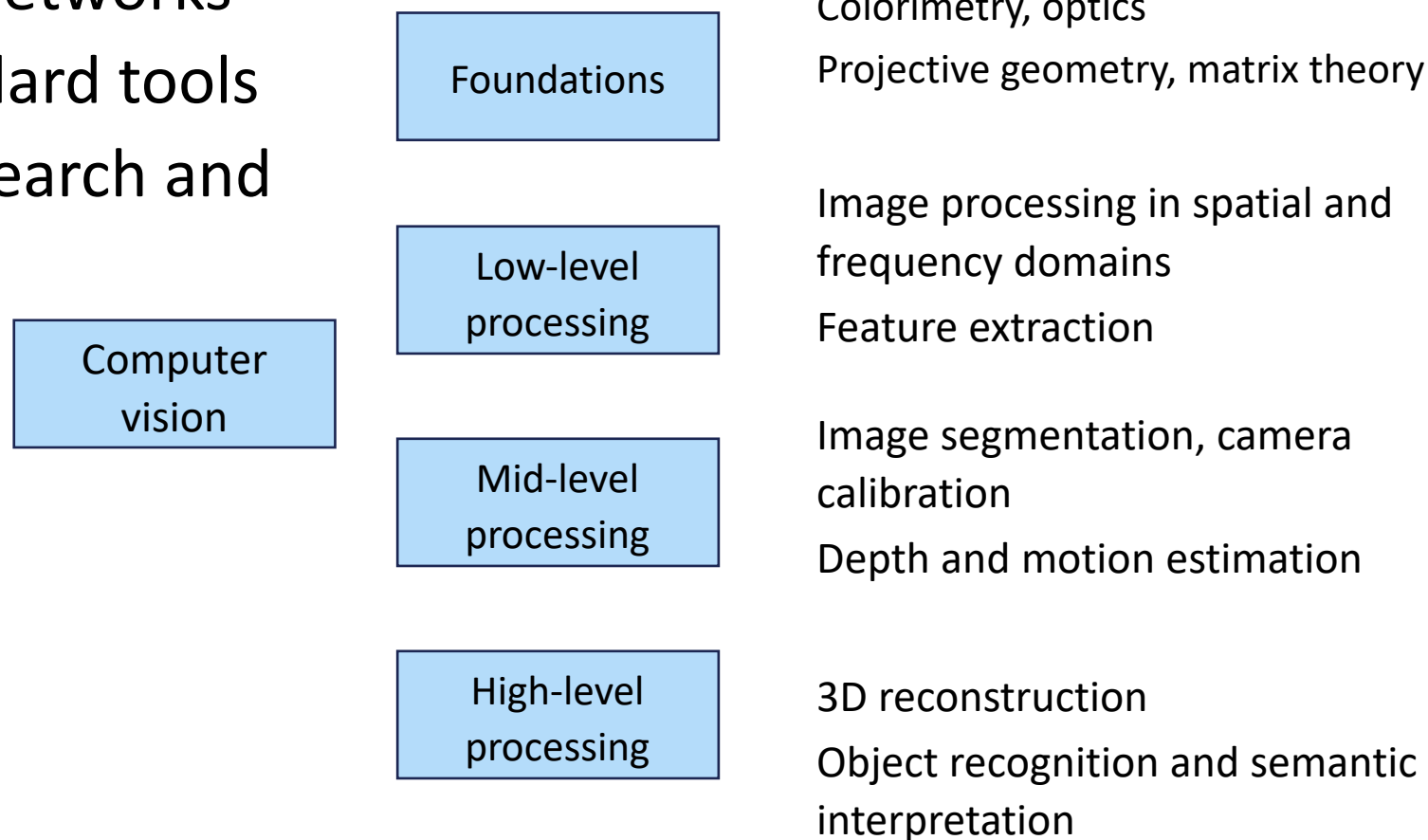
Source: Chen (2005), Figure 5, p. 574. doi:10.1080/13506280444000256

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- **Convolutional Neural Networks**

Convolutional Neural Networks

- Convolutional neural networks are currently the standard tools in computer vision research and applications.



Convolution operation (2-D)

6	0	0	8	4	9
1	4	4	5	0	9
3	9	9	5	8	3
9	9	3	2	5	6
5	9	4	8	5	0
1	3	8	9	0	8

6×6 image

$$6\frac{1}{9} + 0\frac{1}{9} + 0\frac{1}{9} + 1\frac{1}{9} + 4\frac{1}{9} + 4\frac{1}{9} + 3\frac{1}{9} + 9\frac{1}{9} + 9\frac{1}{9}$$



* $\frac{1}{9}$

1	1	1
1	1	1
1	1	1

3×3 filter

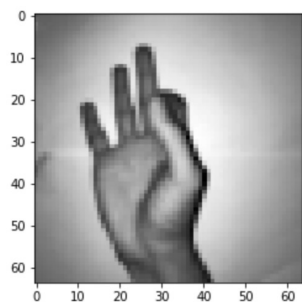
=

4	4.9	4.8	5.7
5.7	5.6	4.6	4.8
6.7	6.4	5.4	4.7
5.7	6.1	4.9	4.8

4×4 output

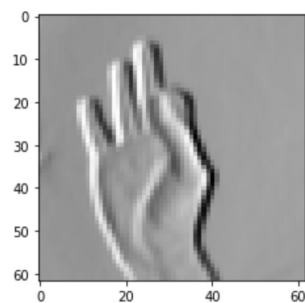
Hand coded filters

Input



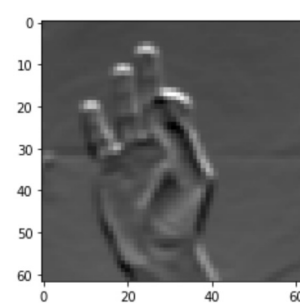
Sobel filter
(vertical)

1	0	-1
2	0	-2
1	0	-1



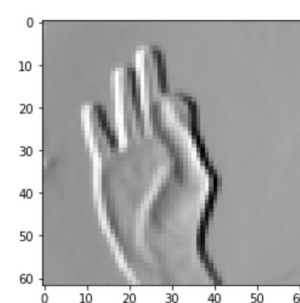
Sobel filter
(horizontal)

1	2	1
0	0	0
-1	-2	-1



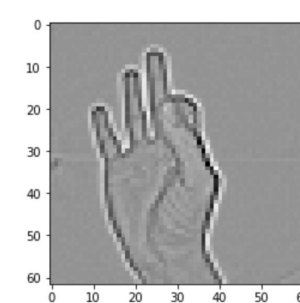
Scharr filter

3	0	-3
10	0	-10
3	0	-3



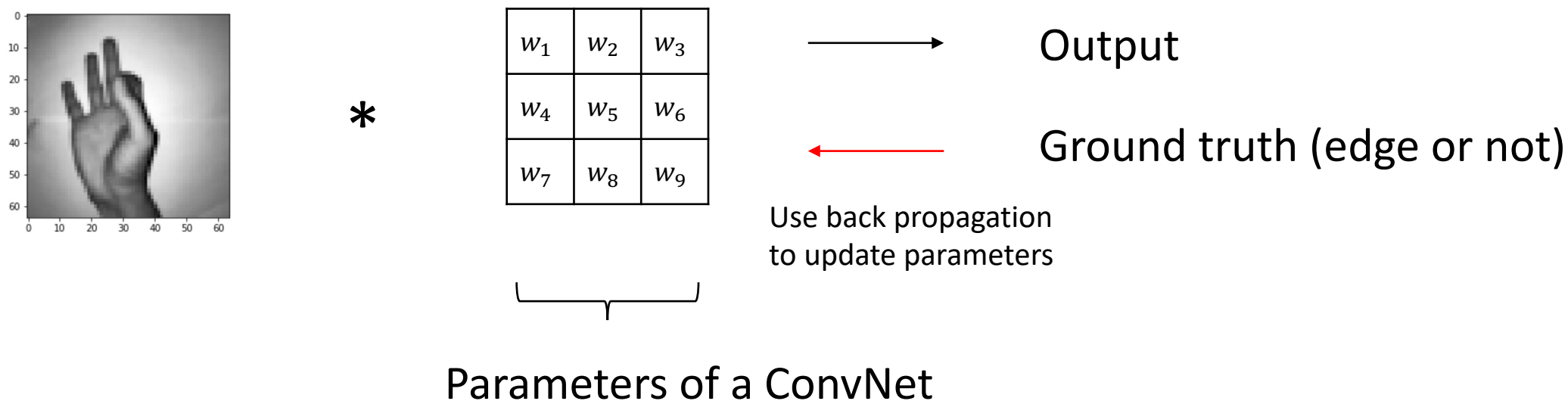
Laplacian
filter

0	-1	0
-1	4	-1
0	-1	0



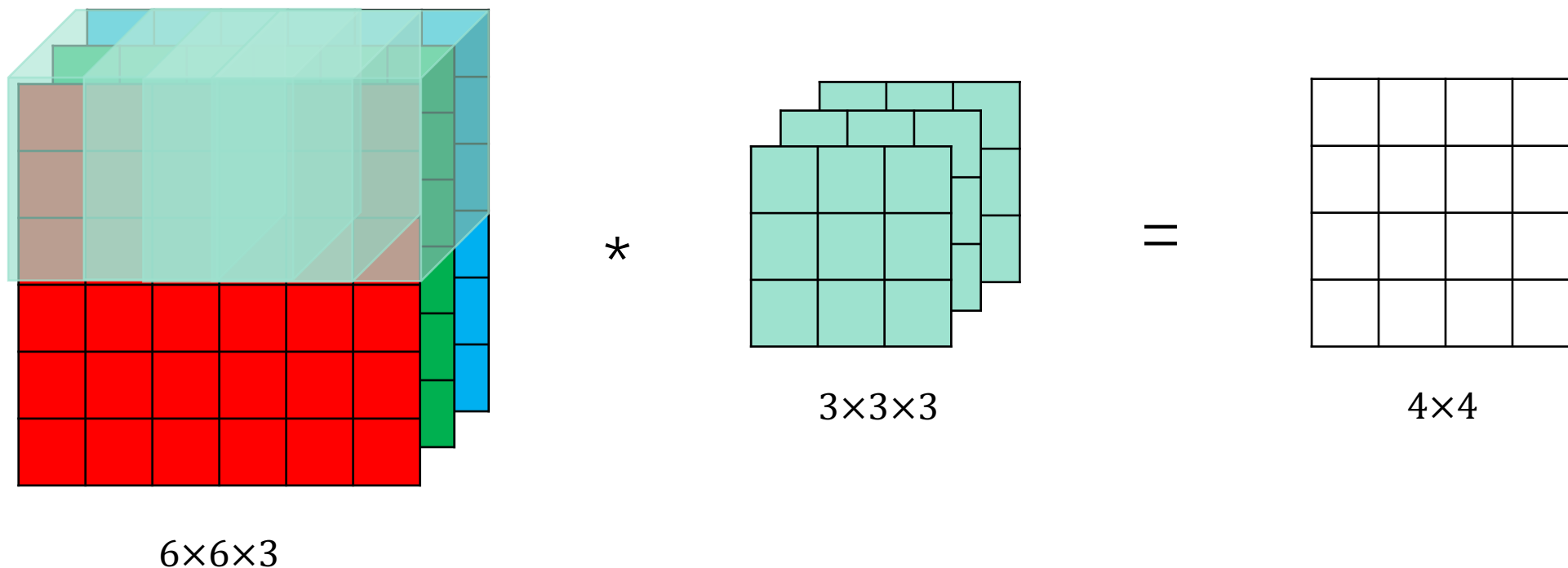
The idea of convolutional neural network

- Not using pre-defined filters, but learning the *parameters* of filters

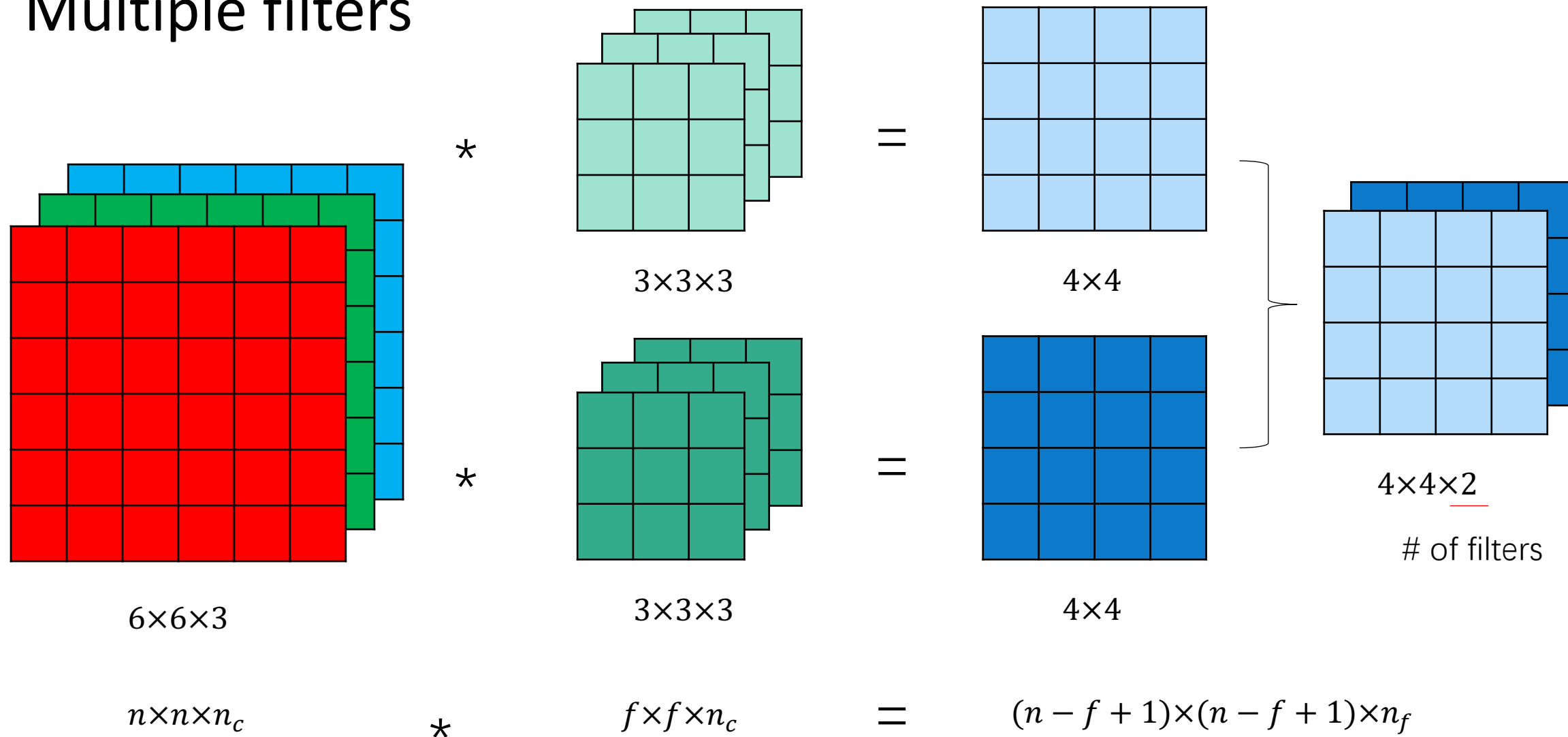


Convolution over 3-D volumes (cont.)

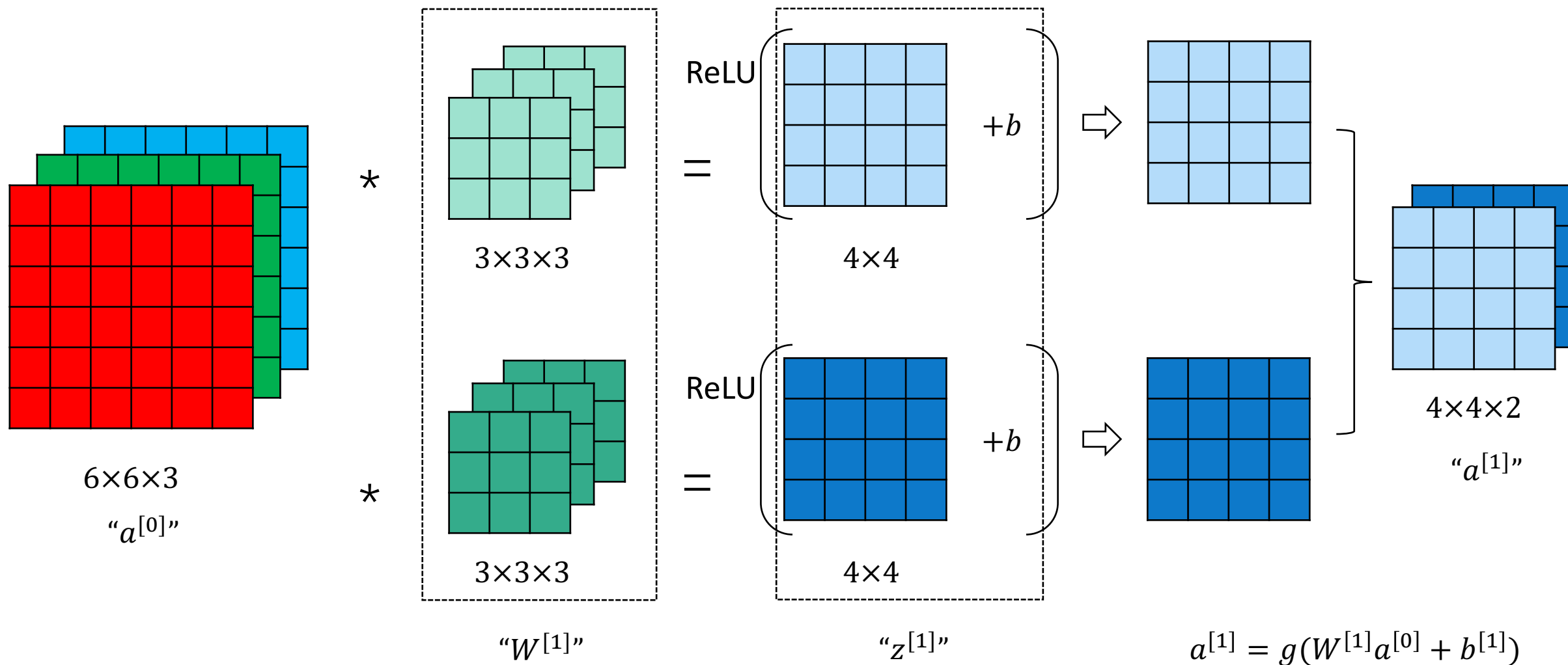
- Element-wise multiplication and then sum



Multiple filters



One layer of ConvNet

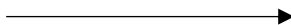


Pooling layer: Max pooling

Input

8	6	3	2
3	10	4	4
5	2	1	1
3	7	2	3

$$f = 2$$
$$s = 2$$



Output

10	4
7	3

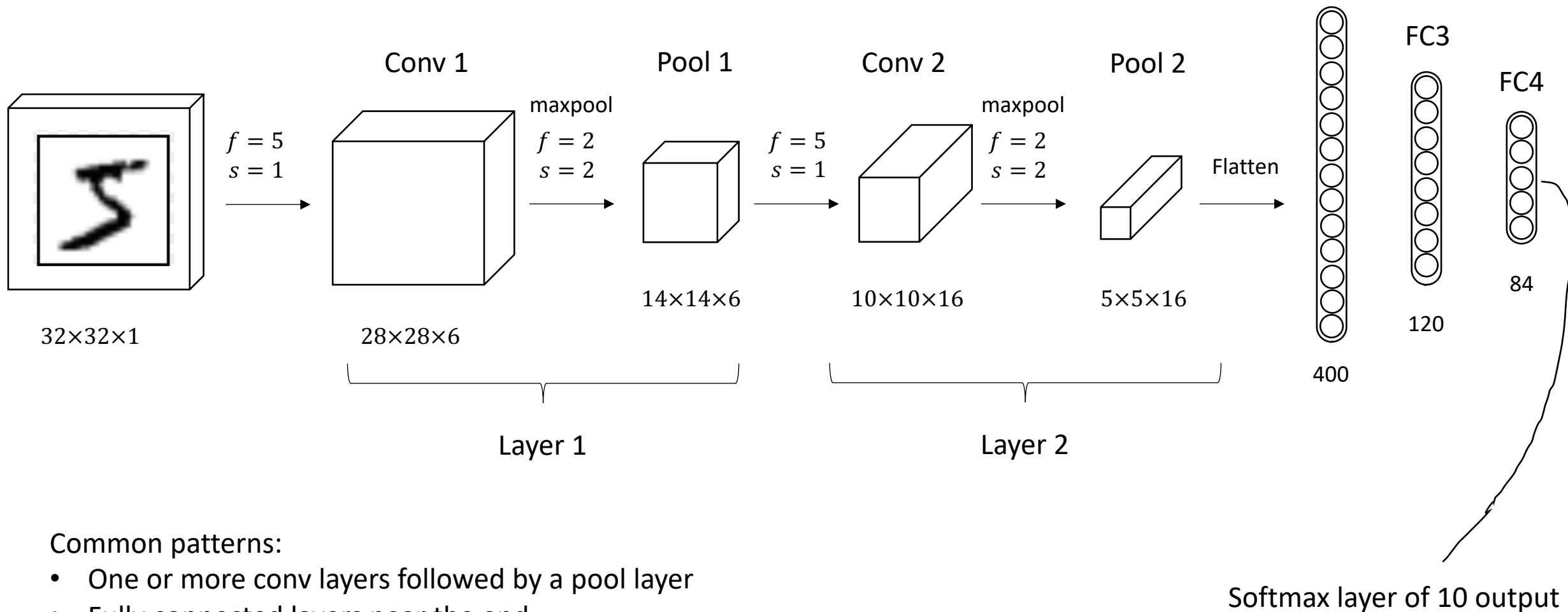
Reduce the size of
representation

Max pooling has no
parameters to learn

Intuition: preserve the features detected

A complete example (variant of LeNet-5)

Original work: Yann LeCun (1998)

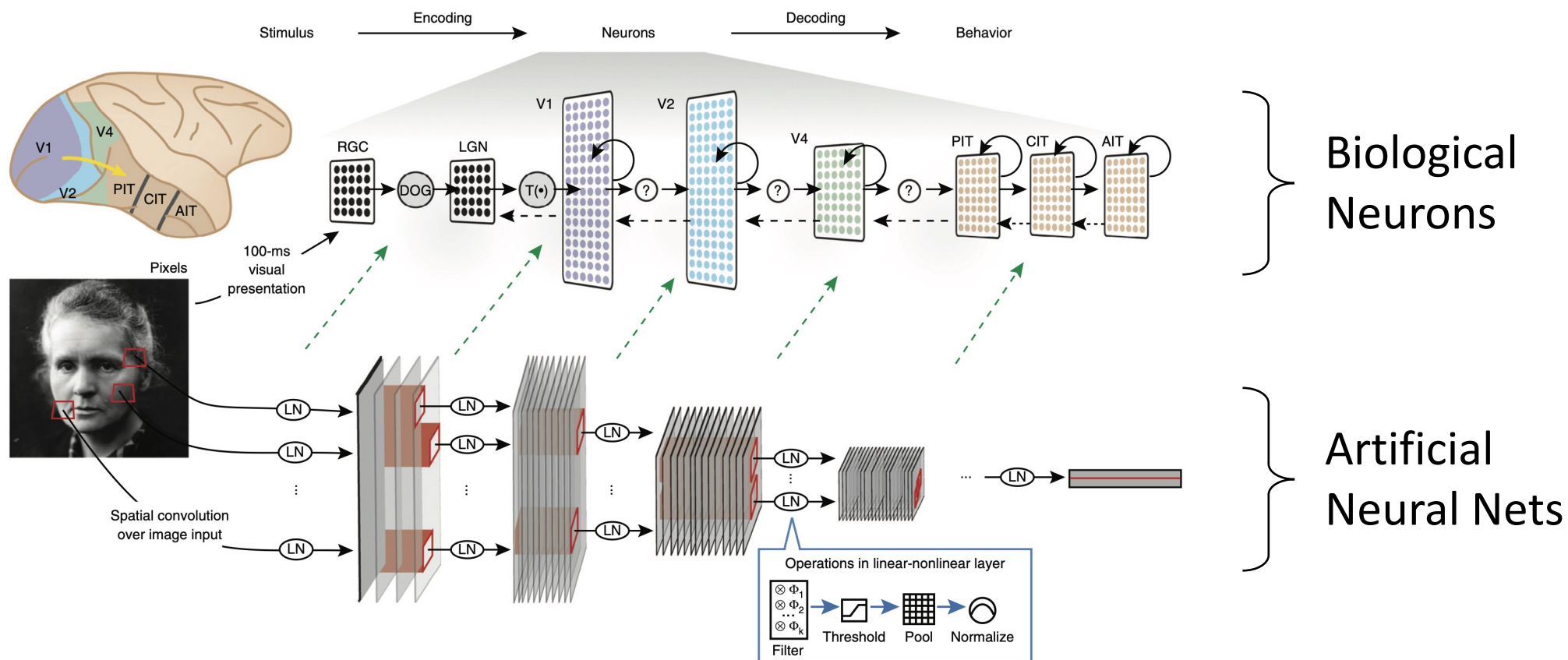


Common patterns:

- One or more conv layers followed by a pool layer
- Fully connected layers near the end

ConvNets and Visual Systems

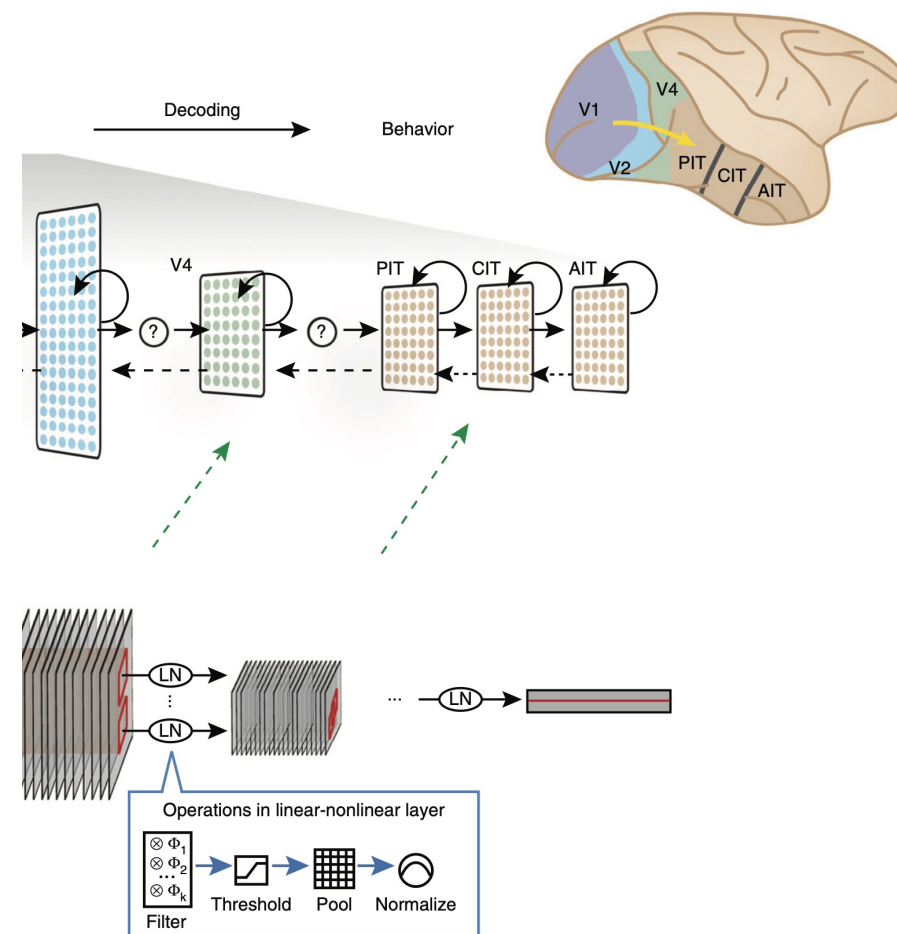
- ConvNets have been proposed as computational models of the visual systems.
(Yamins and DiCarlo, 2016)



ConvNets and Visual Systems

Source: <https://www.nature.com/articles/nn.4244>

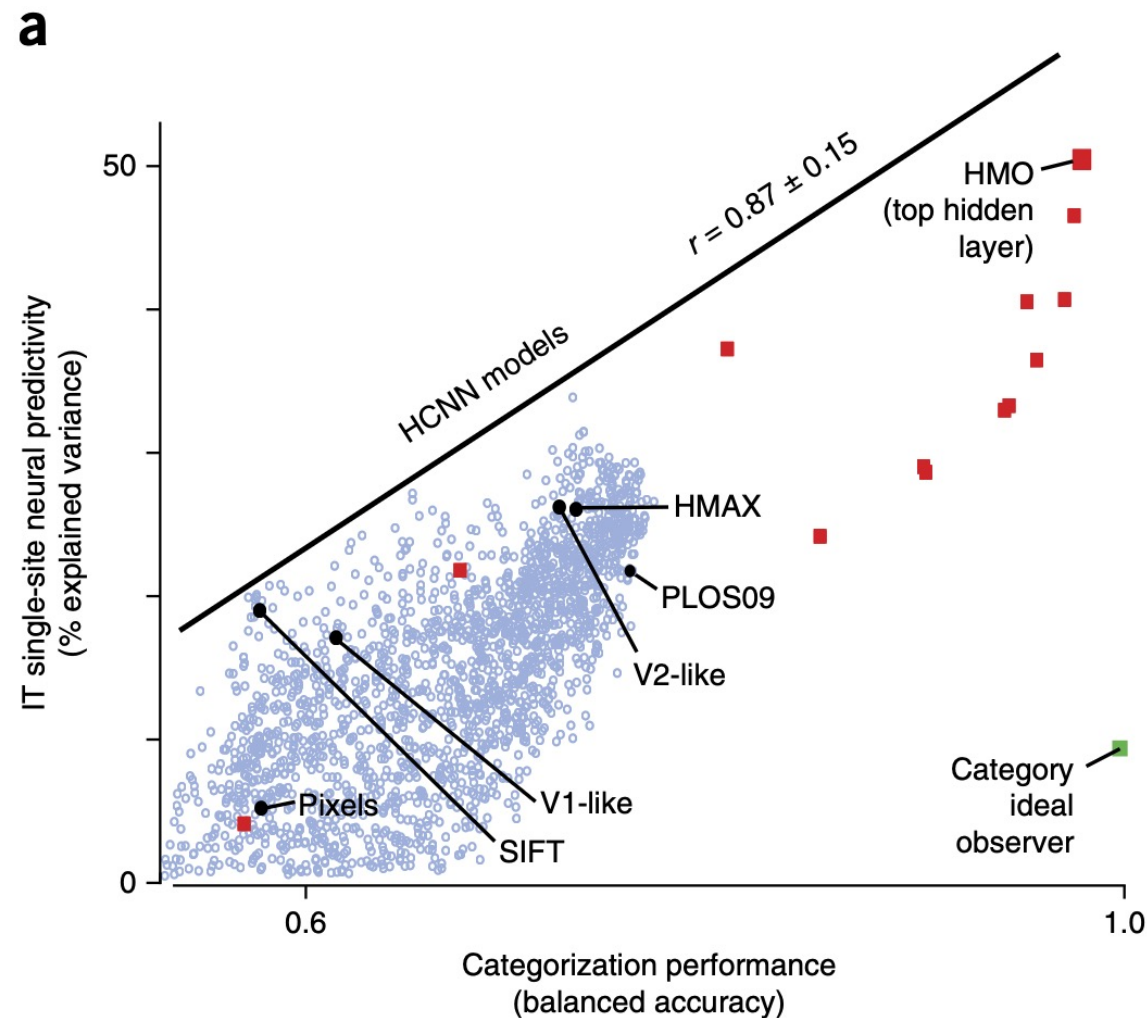
- In Yamins and DiCarlo's (2016) work
- The same image is passed through monkey's visual cortex.
- Neural responses from inferior temporal (IT, 颞下回) is best predicted by responses from the final layer of ConvNets
- Neural responses from V4 is better predicted by an intermediate layer.



ConvNets and Visual Systems

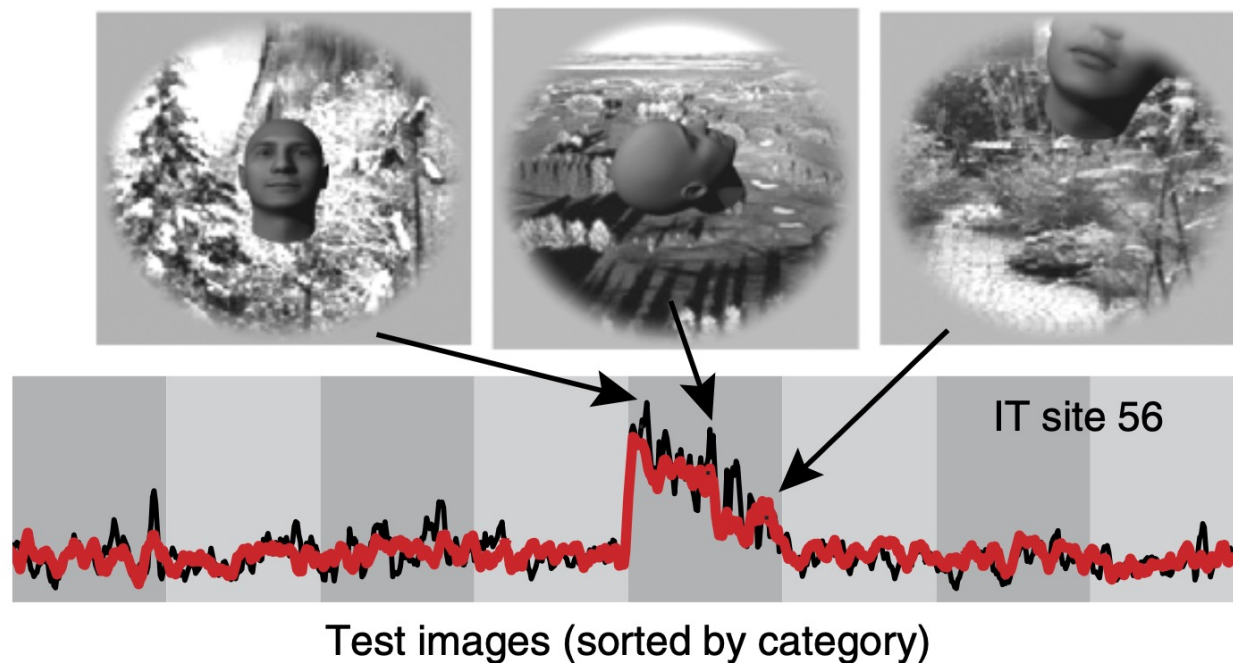
Source: <https://www.nature.com/articles/nn.4244>

- ConvNets models that are better optimized to solve object classification (物体识别) produce hidden layer representations that are better able to predict IT neural response variance.
- y axis: single-site IT response predictability of the last hidden layer of a CNN model.
- site response: mean firing rate 70-170 ms after image onset.
- Each dot represents a CNN model.



ConvNets and Visual Systems

Source: <https://www.nature.com/articles/nn.4244>

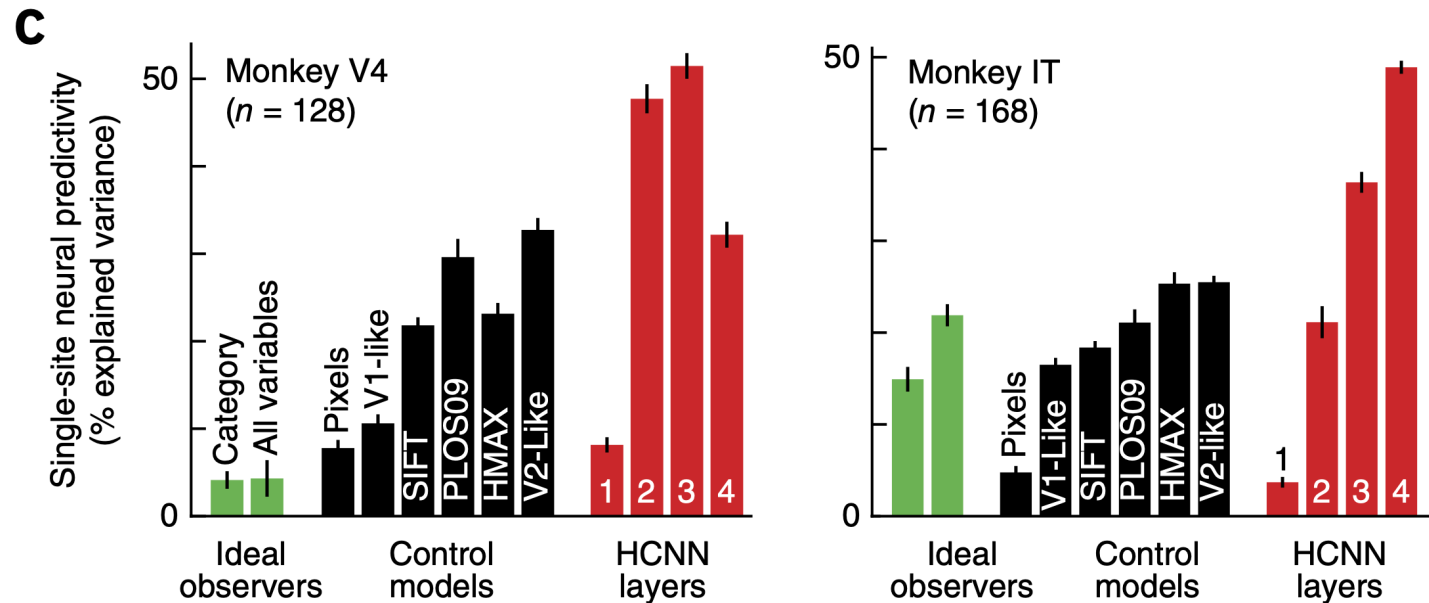


Actual neural response (black) versus model predictions of the last hidden layer of a CNN model (red)

The x axis shows 1,600 test images, **none** of which were used to train the model.

ConvNets and Visual Systems

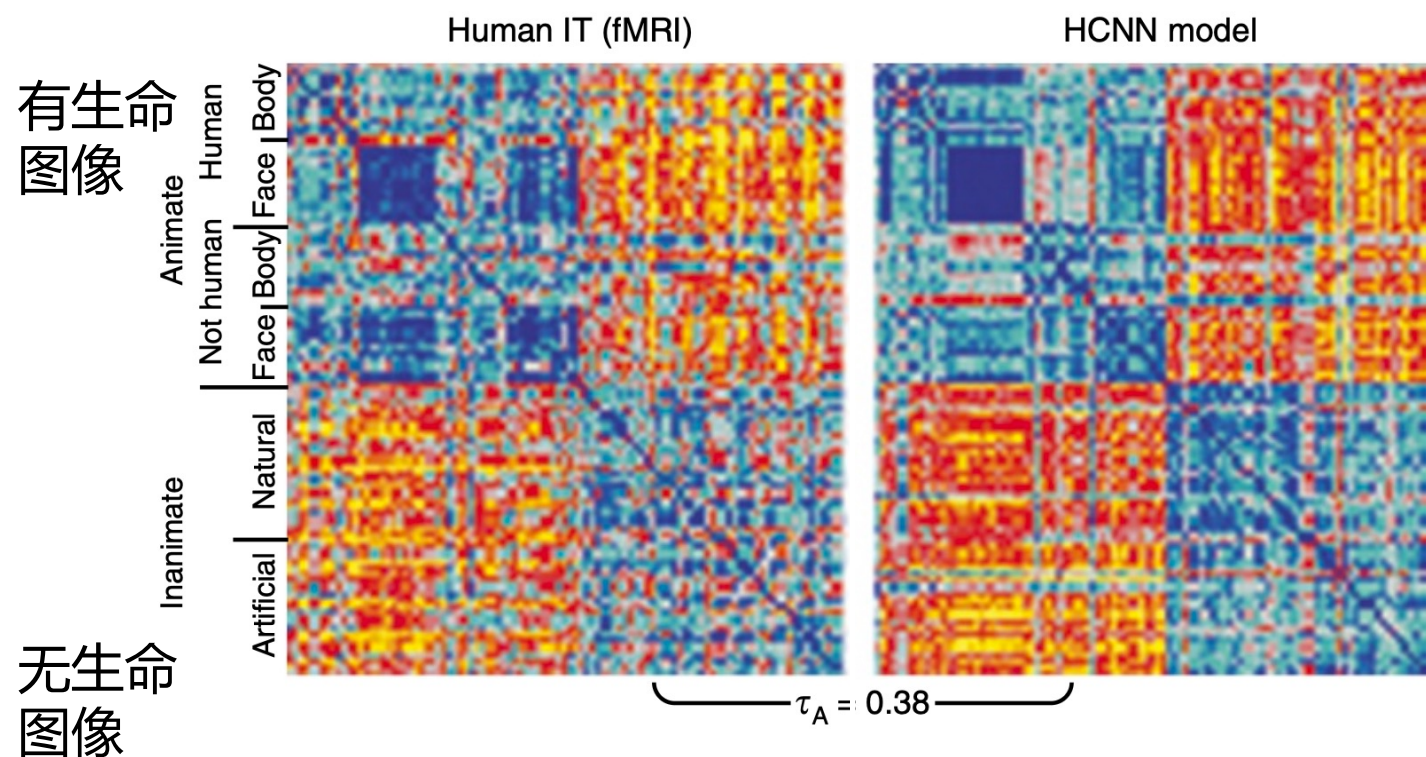
Source: <https://www.nature.com/articles/nn.4244>



The last hidden layer of the HCNN model best predicts IT responses, while the second-to-last hidden layer best predicts V4 responses.

ConvNets and Visual Systems

Source: <https://www.nature.com/articles/nn.4244>



(表征差异矩阵)

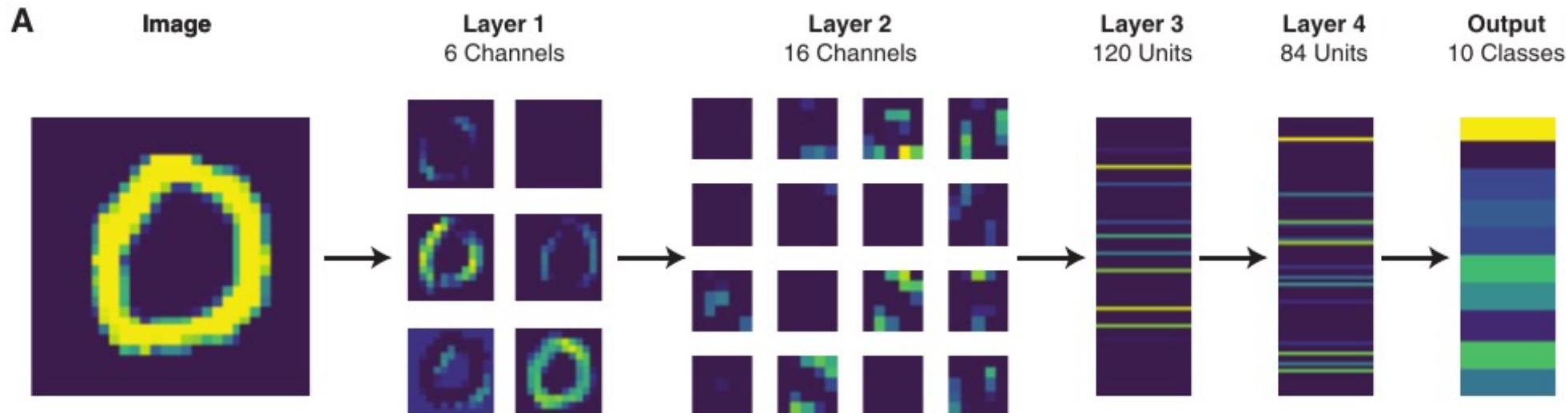
- Representational dissimilarity matrices (RDMs) for human IT and CNN model.
- Blue color indicates low values, i.e., image pairs are similar;
- Red color indicates high values, i.e., image pairs as distinct.

Representational Similarity Analysis (RSA) (表征相似性分析)

How similarly do two systems organize the same images?

Compare which images evoke similar response patterns.

The systems may have different numbers of units.



Source: Yang & Wang (2020), Fig. 4A; Kriegeskorte et al. (2008).

From activations to response patterns

Response matrix R: n images $\times$ p features

Readout	Activation shape	p
l1	$n \times 6 \times 14 \times 14$	1,176
l2	$n \times 16 \times 7 \times 7$	784
l3	$n \times 120$	120
l4	$n \times 84$	84
y	$n \times 10$	10

One row = one image
One column = one feature

Flatten the channel and spatial dimensions.
Keep the image order fixed.

`R = activity[layer].reshape(n, -1)`

Source: ConvNets_origin.ipynb, network definition and representation-similarity cells.

Building a dissimilarity matrix

$$D_{ij} = 1 - \text{PearsonCorr}(r_i, r_j)$$

Response patterns (toy example)

Image	Unit 1	Unit 2	Unit 3
A	1	2	3
B	2	4	6
C	3	2	3
D	3	2	1

Example: $r_A = [1, 2, 3]$, $r_B = [2, 4, 6]$

`np.corrcoef(rA, rB)[1,0] = 1.0`

RDM (Representational Dissimilarity Matrix)

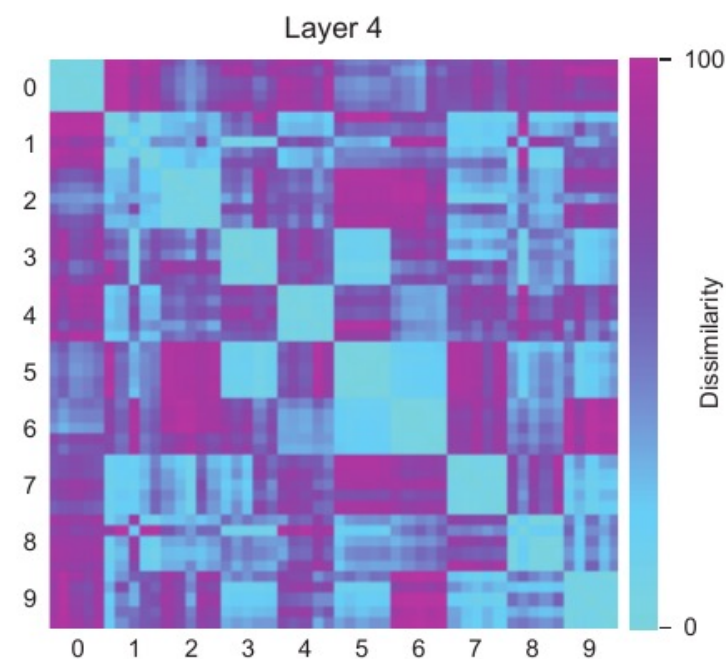
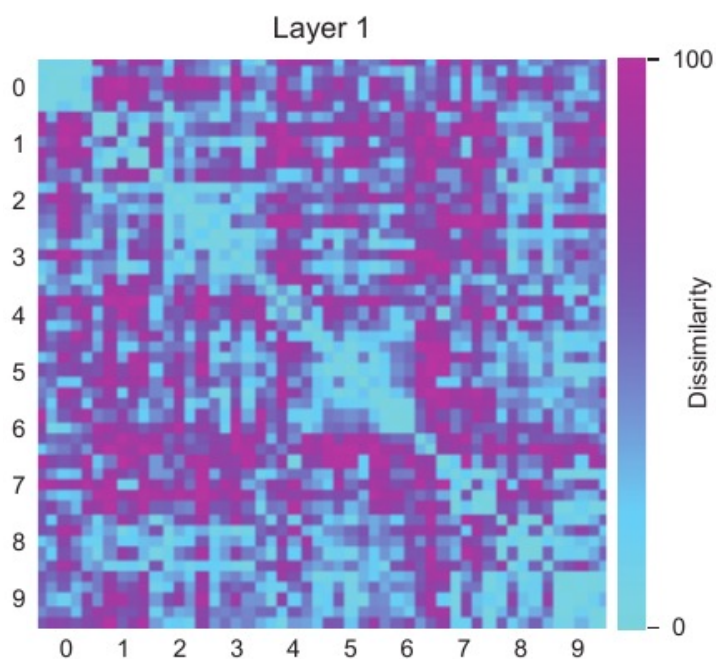
	A	B	C	D
A	0	0	1	2
B	0	0	1	2
C	1	1	0	1
D	2	2	1	0

D is $n \times n$, symmetric, with a zero diagonal.

0: perfectly correlated 1: uncorrelated 2: perfectly anticorrelated

Reading an RDM

Both axes contain the same images, in the same order.



Cyan = lower dissimilarity
Magenta = higher dissimilarity

Cyan blocks near the diagonal indicate similar responses within a digit class.

50 images: five examples of each digit, ordered 0–9.

Source: Yang & Wang (2020), Fig. 4B; percentile display in ConvNets_origin.ipynb.

Comparing two RDMs

$$\text{RSA score} = \text{SpearmanCorr}(\text{upper}(D^A), \text{upper}(D^B))$$

Keep one copy of each image pair

	A	B	C	D
A	—	0	1	2
B	—	—	1	2
C	—	—	—	1
D	—	—	—	—

1. Use the same images and order.
2. Exclude the diagonal and duplicates.
3. Correlate the two pairwise-distance vectors.

Higher scores indicate more similar representational geometry.

Pairs: AB, AC, AD, BC, BD, CD

$$vA = [0, 1, 2, 1, 2, 1]; \quad vB = 0.5 \times vA; \quad \text{RSA score} = 1$$

Source: Yang & Wang (2020); Kriegeskorte et al. (2008). Toy values shown.

RSA in the MNIST lab

Existing notebook

Use a fixed set of test images.
Read out l_1 , l_2 , l_3 , l_4 and y .
Build one RDM for each layer.

```
R = activity[layer].reshape(n, -1)
D = 1 - np.corrcoef(R)
```

Next comparison: digit categories

Category RDM: 0 for the same digit;
1 for different digits.
Which layer best matches this structure?

```
from scipy.stats import spearmanr
Dcat = (labels[:, None] !=
        labels[None, :]).astype(float)
i, j = np.triu_indices(n, k=1)
rho = spearmanr(D[i, j],
                Dcat[i, j]).statistic
```

Checks: identical image order; finite distances; no constant response rows.

Test trained vs. untrained networks. Category agreement is not classification accuracy.